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Learning for Crisis: Improving food security in Uppsala County through participative localized food production

Juan José Ortiz Cuadra

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Uppland region through participative localized
Food production

Juan José Ortiz Cuadra

Supervisor: Nadarajah Sriskandarajah

Subject Reviewer: Cristián Alarcón Ferrari

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Abstract:

An unprecedented situation composed by worldwide land degradation, the post pandemic era, and the Russian invasion of Ukraine is unfolding, reminding us of the impendent risks that phenomena such as these have for food security. Considering that this context presents new unforeseen challenges, and that there is a lack of knowledge production connecting food security to countries of the global north, such as is the case of Sweden, this thesis seeks to foster discussions on food security through the lens of Uppsala County, integrating notions of Local food systems. During the research process, I take on the role of practitioner of Soft Systems Methodology to address the problematic situation affecting Food Security in Uppsala County in a twofold target, which is first to understand the problem and then devise adequate steps for improving the situation, giving essential importance to stakeholders. For achieving this, I performed a quantitative review of statistical data, as well as, qualitative semi-structured interviews to consumers and producers. These were the first steps to comprehend the problem and its possible manifestations. Then for devising plausible actions, I engaged in participatory activities, which complemented the previous results. The findings reveal key vulnerabilities that are interlinked and required further scrutiny by future researchers. At the same time, local food systems, and localizing food production presents in stakeholder's perspective and important road to explore in order to strengthen food security.

Keywords: Food security, Local food systems, Local food, Soft systems methodology, Uppsala County, Sweden

Juan José Ortiz Cuadra, Department of Earth Sciences, Uppsala University, Villavägen 16, SE- 752 36 Uppsala, Sweden

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Summary: Worldwide land degradation, the post COVID-19 era, and the Russian invasion of Ukraine are part of an unfolding situation without precedents. It is reminding us of the implications that events such as these have for food security. Considering that this context presents new unforeseen challenges, and that there is a gap in scientific research regarding food security and countries of the so-called developed world, such as is the case of Sweden, my thesis seeks to foster discussions on this issue, linking them to the specific context of Uppsala County, and integrating notions of local food systems. For guiding my investigation, I rely on Soft Systems Methodology to address the problematic situation affecting Food Security in Uppsala County. My research aim had a twofold target, which was first to understand the problem and then devise adequate plausible measures for improving the situation, giving essential importance to the people concerned about the situation. For achieving this, I performed a statistical data review, as well as, interviews to local consumers and producers. This as part of the first phase that was to understand the problem and its possible manifestations. Then for devising actions, I engaged in participatory activities, which complemented the previous results. The findings reveal some important vulnerabilities that are interrelated and required further research. At the same time, localizing food production provides in the participants perspectives and important road to explore in order solve the problematic situation.

Keywords: Food security, Local food systems, Local food, Soft systems methodology, Uppsala County, Sweden

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1. Introduction

We are living in a time where the combined effects of continued land degradation, the post-COVID-19 era and the ongoing Russian invasion of Ukraine have reminded us in the severest manner possible of the inherent, interconnected implications that a wide range of stressors such as these have for the global Food System. (EU commission, 2022; World Bank, 2022).

First, on a global scale, climate change has already affected the four so-called pillars of food security: availability, access, utilization, and stability (FAO, 2022). Furthermore, studies conducted by the IPCC (2022) reveal that this destructive pattern will escalate in the near future, affecting crop yields, livestock systems, supply chains, production costs, food prices, and safety. If the current trajectories of climate change persist, it will lead to losses in rural livelihoods, income and ecosystems (Rockström et al., 2009; FAO, 2015), that will intensify deterioration of food security.

In the case of the COVID-19 pandemic, it shocked food systems in multiple dynamics. Due to lockdowns impositions, trade was restricted, and supply chains suffered major disruptions. In turn the economy decelerated, which resulted in heavy losses of income, rise in agricultural inputs and food prices, thus affecting household access and availability to food (Clapp *et al.*, 2022). After three years, even if the critical stage of the pandemic is over, the post-effects relating to food security will linger on (McDermott & Swinnen, 2022). According to studies performed by the British Academy (2021), the incrementing unemployment will evidenciate this, by widening food poverty and malnutrition in marginalized groups for years to come. In addition, the Russian invasion of Ukraine is having a cascading impact, further disrupting key markets and supply chains for crops and fertilizers that significantly affect global food security (World Bank, 2023).

The amalgamation of these events has unleashed an impendent phenomenon that is perturbing the different regions of the world in disparate ways. Latin America, the Middle East, Africa and South East Asia have been and will be the most damaged by it (FAO, 2022). In correspondence, the majority of recent and current studies direct efforts to investigate and address issues in these regions. This has left research gaps in parts of the world that are often called "The Global North" and which are deemed to be food secure or less vulnerable than the aforementioned regions (Horn, Ferreira and Kalantari, 2022; Borch and Kjærnes 2016). What is clear is that in the actual context, many countries that have been indexed as food secure have now become vulnerable and are facing new challenges (Larsson & Sjöqvist, 2022). Especially susceptible are net importers of food that are indirectly exposing themselves to disturbances occurring beyond their own borders (Cottrell et al. 2019; Kummur et al. 2020). A clear example is the case of Sweden, where studies conducted by Rost and Lundälv, 2022 expose that from 1697 articles scrutinized, only five related to the topic of food security in Sweden.

Understanding that there is a need for additional studies that discuss on the augmenting vulnerabilities that threaten food security in the "Global North", this Thesis seeks to contribute by learning from the situation, and provide inputs that foster these discussions by addressing the case of the Uppland region in Sweden. My work builds on the foundations of local food systems as an alternate path to transcend the dominant approach to Food Security. Additionally, to signal vulnerabilities and potential solutions that may reinforce preparedness in the face of impending crises.

1.1. Problem Formulation

Considering that Uppland is embedded in the larger Swedish and Global Food System, it is prone to the worldwide disturbances inflicted by COVID-19, the Russian invasion of Ukraine and continued landscape degradation. These phenomena present a problematic situation that puts pressure on food availability, accessibility, utilization or stability (Savary et al., 2020). In this light, Swedish authorities decided to design a plan to learn from these crises, improve food supply and heighten preparedness (Jordbruksverket, 2022). Unfortunately, there is still no clear target for Sweden's and its regions

capabilities to secure food. Moreover, the measures focus mostly on the supply chain, market mechanisms, trade routes and increased food production (Jordbruksverket, 2022); overlooking significant indicators that could provide an estimate of the current food situation. In addition, there is no pathway for enhancing supply chains by localizing food networks and production. An illustration is Sweden's high reliance on international trade and dependency on food imports, mostly fruits and vegetables (75% Jordbruksverket, 2021). This reveals a notable vulnerability to disruptions in its Supply chain that need to be rethought and addressed. Furthermore, even if the devised plan is a "good declaration of intentions", there is no clear framework to learn and understand the complexity of an unprecedented situation that can be deemed problematic in several aspects. The aforementioned factors combined with a neglecting of civil involvement in policymaking, increased prices in local agricultural production and geopolitical and climate uncertainties compose the ideal "ingredients" for an impendent food insecurity that will reverberate through all scales of the Swedish context. (Horn, Ferreira and Kalantari, 2022). From this stance it is noticeable that there is a dearth in learning and knowledge production that needs to be addressed.

1.2. Aim and Research Questions

The overarching objective of this work is to provide an integral discussion on food security through the lens of Uppland Region, integrating notions of local food systems. This as a first step to motivate constant learning from the problematic situation, as well as, co-produce knowledge by fostering space for dialogue concerning these issues in Sweden. For achieving this, the aim of this thesis is to address the problematic situation concerning food security in the region caused by worldwide land degradation, the post pandemic era, and the Russian Invasion of Ukraine, and. In concordance, a twofold target guided by Soft Systems Methodology and informed with Systems Thinking is set. Consequently, first, I seek to understand what the problem is, as well as its possible manifestations in terms of risks and vulnerabilities. Then devise adequate steps for improving the situation. In addition, to enrich the process with the participatory element, paramount importance will be committed to the perceptions of consumers and producers in the region as essential for informing this research.

Based on the problem formulation and aim, the following research questions have been formulated:

- 1) *What vulnerabilities exist is the current situation of food security in Uppsala County?*
- 2) *To what extent can food security improve by localizing production in Uppsala County?*

1.3. Positionality

Positionality as explained by Holmes (2020) becomes an important aspect that has an impact on my bias as Researcher. For instance, my preconceptions may influence the selection of interviewees and the interpretation of their responses. In this light as proposed by and Savin-Baden & Major (2013), it is important to identify one's positionality, including personal beliefs and assumptions about social reality. In my case, I am a 36-year-old, middle class Nicaraguan white Latin male, engineer, agnostic atheist and without any particular politic or ideological affiliation. These features have an influence on how I understand the world, and in how I comprehend a particular research process, thus shaping my positionality. Exercising self-reflection, I can express that probably the trait that affects most my positionality is my experience in politics and social organization. I believe that I grasp the main dynamics of power relations and power structures, either in Nicaragua or in any country (also at different scales of power, or organization levels). This experience tilts my positionality towards the belief that most or all, dominant discourses have the characteristic of being proposed or enabled by groups or organizations that have power or are seeking power. This has the effect on me, to try to situate myself outside any discourse. This does not mean that I am at always at conflict, but that I tend to try in a utopic way to bring together, mediate, and conciliate ideas from different discourses, even contrasting worldviews, instead of completely discard one or fully follow another.

2. Background

2.1. Food security as a concept

Food security as a concept has changed over the past five decades. Some of the first discussions occurred during the early 1970's and, it was first defined during the 1974 World Summit as "availability at all times of adequate world food supplies of basic foodstuffs to sustain a steady expansion of food consumption and to offset fluctuations in production and prices" (FAO, 1974). In 1983, FAO expanded its concept to incorporate securing access by vulnerable people to available supplies. The definition was then renovated as "ensuring that all people at all times have both physical and economic access to the basic food that they need" (FAO, 1983). Consequently, the report "Poverty and Hunger" elaborated in 1986 by the World Bank broadened the reach of the concept by considering the dynamics of Food Security in relation to chronic and transitory food insecurity (Clay, 2003). Again, the term was updated to "access of all people at all times to enough food for an active and healthy life (WB, 1986). During the 1990's, as worldwide awareness and preoccupations towards hunger grew, the definition was modernized by transcending the traditional focus on availability, now addressing issues of nutritional values and safety of the aliments (Maxwell, 1996).

The current and most commonly used definition was coined during the 1996 World Food Summit. Food security became more complex, identifying five levels or scales; individual, household, national, regional and global. With these differentiations, food security was agreed to have been achieved when "all people, at all times, have physical and economic access to sufficient, safe and nutritious food to meet their dietary needs and food preferences for an active and healthy life" (FAO, 2006). This actualization of the concept meant a key shift towards including a household and individual perspective in contrast to only a global and national point of view, assessing food but also livelihood conditions, and acknowledging subjective perceptions such as food preferences (Maxwell, 1996). Since then two important enhancements have been made. First, after the State of food insecurity summit in 2001, FAO revised to "a situation that exists when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food that meets their dietary needs and food preferences for an active and healthy life" (FAO, 2002). Secondly and most crucial food security was provided with a set of "dimensions" (access, availability, utilization, stability) (FAO, 2006) that are known as the four pillars. These aspects facilitate assessment of food security by locating the focus and production of indicators in relation to them (Clapp et al., 2022)

2.2. Sweden in the context of food security

According to Rost and Lundälv (2021), food insecurity is not a term usually associated with a well-established welfare state such as Sweden. Most Swedes frowned when a relation between the country and food insecurity is insinuated. Moreover, there seems to be a notion that such topic is foreign to its borders and delimited to other regions of the world. Despite this, European countries and Sweden is no exception, are vulnerable to the incrementing risks that damage food systems.

Even before the coinage of the concept of food security, Sweden possessed a strong contingency plan as part its civil defense. After both world conflicts, and during the cold war, food supply and accumulation of agricultural inputs became top priorities for achieving high levels of self-sufficiency. Nonetheless, after the dismantling of the Soviet Union, and the subsequent incorporation of Sweden to the EU, its contingency system in light of this new Era expired. (Eriksson, Fischer & Ulfbecker, 2020). This translated into an unnecessary obligation of storing food or agricultural inputs; its supply now rests on trade mechanisms that respond to the Global or regional EU market (Eriksson, 2018).

However, in the actual context, the coexistence of the triple threat of worldwide land degradation, the post-pandemic era and the Russian invasion in Ukraine has begun to stimulate research and discussions in the Swedish community where concerns on the country's approach to food security are at the core. First, as a measure to tackle the effects of unpredictable land degradation, the government released a

new defense act in 2015. This targeted for an increased budget in relation to improve planning in aspects such as food security (Swedish government, 2015). In addition, The National Food Strategy, introduced in 2017, aimed to increase the self-sufficiency of the country. The approved bill is one of the first official documents to have revived notions on incrementing domestic food production. (Swedish government, 2017). According to Jordbruksverket (2021), the plan aimed to be a long-term growth strategy for the entire food system, which extends until the year 2030. Additionally, to develop a competitive food chain where the total food production increases, without compromising environmental goals.

Secondly, because of the afflictions caused by COVID-19, the Swedish Agency for Agriculture and the Swedish food agency produced in spring 2021 the report “A robust food supply in crises and heightened preparedness”. The document analyzed the impact of the pandemic on the food supply and helped to facilitate dialogues within Government offices for outlining “Sweden’s Pathway for Sustainable Food Systems”. These discussions pointed out that the Sweden has still many weaknesses that need to be address focusing on systemic and integral approaches.

Moreover, currently after Russia’s invasion of Ukraine the Swedish authorities are working on revisions to the country’s proposal of the Common Agricultural Policy (CAP) strategic plan. This initiative, which is part of a regional shared stratagem for accomplishing sustainable agriculture in the EU, has become an important roadmap for strengthening preparedness and reducing vulnerabilities from previous national policies. This document under the assessment of the EU, also presents its importance by revealing several aspects in Sweden’s proposal that require enhancing in order to address the actual context of impending crisis (EU commission, 2022).

Conclusively, what needs to be understood from all the aforementioned initiatives, discussions, plans, reports is that further civil involvement needs to exists, combined with greater domestic production of certain goods (FAO, 2022) (such as fruits and vegetables), with an emphasis on local small farmers and diversified products. To complement, in the risk of difficulties accessing agricultural inputs, also organic cultivation needs to increase (EU commission, 2022).

2.3. Vulnerabilities of Sweden’s food supply

A preliminary scrutiny of Sweden’s food systems shows that vulnerabilities are entangled in a complex web of risks that relate to the country’s food supply. This comes from Sweden’s historical dependency on food imports due to seasonality and weather conditions. This becomes relevant because as Cottrell et al. (2019) and Kummur et al. (2020) highlight, countries in the global north that are net importers of food are more vulnerable to experience food insecurity. This complicates the situation further, as there are risks and vulnerabilities related to the unique contexts, countries or region where specific products are coming from. These are highly difficult to assess and predict. A clear example is avocado production in the Petorca region in Chile, which traditionally has been certified as sustainable. However, research has exposed that its cultivation for exportation to countries such as Sweden has provoked several socio ecological issues, such as desertification, privatization of water and human rights violations (Wehr, 2018). This implies an obstacle for achieving sustainable pathways for Sweden’s food system manifesting a probable incapability of managing the risks associated to food supply (Antonissen, 2020).

Reviewing the situation a little deeper is possible to notice a risk concerning supply chain stability and Sweden’s necessity of food imports. Especially critical is the proportion of fresh fruit and vegetables being imported which by 2022, minimized domestic production. At the same time, consumption of vegetable and fruits has increased 4% in the past two years. Depending on the product, there is a 100% dependency on imports from other countries (Jordbruksverket, 2022). This also hints for a probable scarcity in certain products if supply chain or market is affected. Scarcity that is becoming more tangible as consequence of conflicts around the world, especially the war in Ukraine. The spike in fuel and agricultural inputs prices have increased production costs derived from this situation, which in turn affected the capability for companies and retailers to cope with the actual

demand due to supply chain disruptions and higher costs for transportation and logistics (EU commission, 2022; (Ben Hassen & El Bilali, 2022). Traces of these complications can be examined in reports from FAO (2015, 2020, 2022) and WB (2015, 2020, 2022) which reveal that since 2015, the prevalence of moderate or severe food insecurity in Sweden's population has increased.

2.4. Uppsala County vulnerabilities within the Swedish food context



Figure 1. Map of Uppsala County (constructed after TUBS; 2011, and Portunes; 2018)

Uppsala County (Figure 1.) as part of a broader context is affected by the dynamics and interactions of the larger Swedish food system (Eriksson, Pano, & Ghosh, 2016). A result of this interplay between scales on the global food chain is that, the “insecurity” patterns described in the previous chapter concerning the Swedish food system context are in consequence reflected in the Uppsala region. Elements that are ideal ingredients for a food crisis are combining and increasing vulnerabilities, which leads to new threats and escalating risks for Sweden's and Uppsala food security (EU, 2022).

An important feature that undermines the capacity of local farmers to crop is the escalating costs for agricultural production. For example, the average prices for means of production increased according to the Means of Production Price Index (PM-index) by 20.6% between December 2020 and December 2021. Furthermore, prices for the product group Fertilizers and soil improvers increased the most (+158.4%). To illustrate how these increments translate into a higher crop production cost, it is possible to mention the specific case of Potatoes where prices increased by +48.3% between 2020 and 2021 (Jordbruksverket, 2021).

In addition, another factor that is negatively affecting local food production is a disproportionate rise in agricultural land prices (pastures, arable and agricultural land). To grasp how this can influence farming activities, statistics reveal that since 2010 in the case of pastures, prices are up 149%, arable land has experienced an increment of 126%, and agricultural land 121%. Considering this, it is evident that the high prices in agricultural land and production costs are preventing farmers whether organic or conventional, from further producing locally in Sweden and Uppsala and causing still more dependency on imports (Jordbruksverket, 2022).

3. Theoretical Framework

For this chapter, the main approaches that guided this thesis are explained, as well as their utilization in the research process and justification for their selection. At this point, it is also paramount to express that even if some of the worldviews, theories, or conceptual frameworks may appear contrasting or as opposites, it is my belief that in some cases, certain features of these approaches may be conciliated as part of unique contextual solutions to a situation or problem.

3.1. Systems approach to food security

A systems approach to food security relies on Systems Theory and Systems thinking to constitute an integral framework that can address complex, intertwined food production, and supply issues. In practical terms, it serves as a conceptual framework for research and policy aimed at sustainable solutions for the sufficient supply of healthy food (Van Berkum, Dengerink, & Ruben, 2018). The initial premise is to conceive a global food network, which encompasses an entire range of actors and their interlinked value-adding activities involved in the production, aggregation, processing, distribution, consumption and disposal of food products that originate from agriculture, forestry or fisheries, and parts of the broader economic, societal and natural environments in which they are embedded (FAO, 2018). In congruence with systems thinking, the food system is composed of sub-systems (e.g. farming system, waste management system, input supply system, etc.) and interacts with other key systems (e.g. energy system, trade system, health system, etc.) (IFPRI, 2012).

At the same time, all these interactions, and activities, require inputs, products and services that cause environmental impacts while also being influenced by social, political, cultural, technological, economic and environmental aspects. (HLPE, 2019; UNEP, 2016). Within systems theory, a food system can be considered as a complex system (Mesarovic and Takahara, 2009), as each of its components can be classified as a system on its own.

For Food Security, a system approach allows for a combination of all the factors that are crucial for ensuring it. This also facilitates a better alignment to various problems in the organization, availability and distribution of the current food system along with its effects on people and the environment. For example, malnutrition is inextricably linked to the sufficient production of healthy food, while climate change has an effect on production. (Van Berkum, S., Dengerink, J., & Ruben, R., 2018). In addition, for this Thesis a systems approach is key, due to its feature on identifying different, sub-systems, elements and interactions across scales, as is the case of the Uppland region, which is embedded into larger food systems. This allows for an integral understanding of relation, processes, and functionality between different elements or sub-systems (Checkland, 1984; Meadows, 2008). At the same time is essential to have a systemic approach, as most institutions that engage in the study and policy design related to food Security use frameworks derived from Systems Theory. Some evident examples can be noted in the references; UN, EU, IPCC, OECD, FAO, IFPRI. To complement the importance of a Systemic vista for this work, the problem solving oriented approach chosen as framework for managing risks and potential enhancements is Soft Systems Methodology (SSM).

3.2. Food Security framework

A food systems approach is a useful interdisciplinary conceptual framework for research and policy aimed at sustainable solutions for the sufficient supply of healthy food. It analyses the relationships between the different parts of the food system and the outcomes of activities within the system in socio-economic and environmental/climate terms. A food systems approach sheds light on non-linear processes in the food system, and on possible trade-offs between policy objectives. Systems thinking also broadens the perspective when seeking solutions for the root causes of problems such as supply chains and agricultural inputs. The framework offers at least three benefits. First, it provides a checklist of topics that should at the very least be addressed when it comes to improving food security, in relation to other policy objectives. Second, a food systems approach helps to map the potential impact of variables, such as supply chain disruptions, increased prices, land degradation or conflict on food security by pointing to the various vulnerabilities of the food system. In this sense, the approach can contribute to the search for possibilities for strengthening the system's robustness to disturbances. Third, it helps to determine the most constraining factors for achieving food security, and hence identify effective interventions aimed at improving food security (Van Berkum, Dengerink & Ruben, 2018).

For this Thesis, the food security framework is paramount for finding possible solutions problems reviewing the entire chain, considering the production, processing, distribution and consumption of food as well as sustainability, land degradation and possible external vulnerabilities. More specifically a systemic scrutiny will help to assess the connections and interdependence of people, elements, and organizations across the system in order to: (1) determine system efficiency. (2) Quantify externalities. (3) Understand people's behaviors. (4) Understand the evolution of food systems. (5) Assess the consequence of changes (policy, intervention, or shocks), and (6) identify risk-hotspots and intervention target points (Alarcon et al., 2021).

3.2.1. The four pillars of food security

Four dimensions have been integrated to the conceptual framework of food Security (FAO, 2006). (1)Availability. Refers to the physical existence of food, be it from own production or on the markets. On national level food availability is a combination of domestic food production, commercial food imports, food aid, and domestic food stocks, as well as the underlying determinants of each of these factors (Weingärtner, 2004). (2)Accessibility is achieved when food can reach the consumer (transportation infrastructure) and households along with its individuals having enough economic resources to obtain it (FAO, 2008). To physical and economic accessibility, socio-cultural access is added to ensure that the food is culturally acceptable and that social protection nets exist to help the less fortunate. (3) Utilization signifies that individual must be able to eat adequate amounts both in quantity and in quality in order to live a healthy and full life to realize his or her potential. Food and water must be safe and clean, and thus adequate water and sanitation are also involved at this level. A person must also be physically healthy to be able to digest and utilize the food consumed (FAO, 2008). Cooking habits also need cautious evaluation because some methods of may lead to the loss of essential nutrients (Swaminathan, 2006). (4) Stability defines the temporal dimension of food and nutrition security, individually the period in which food security is being considered. Stability is assumed when the supply food, income and economic resources remain constant in a household level during the year and in the long-term. Besides, it is also essential to minimize external risks such as natural disaster, climate change, price volatility, conflicts or epidemics, etc. (Klennert, 2009). This means that stability, also accounts for the ability of the nation, community, household, or person to withstand shocks to the food chain system whether caused by natural disasters (climate, earthquakes) or those that are man-made (wars, economic crises). Thus, it is possible to observe, that food security exists at a number of levels or scales. Availability - National; Accessibility – Household; Utilization – Individual; Stability. All four of these dimensions must be intact for accomplishing food security. Recent developments emphasize the importance of sustainability, which may be considered as the long-term time fifth dimension to food security. Sustainability involves indicators at a supra-national

and regional level of ecology, biodiversity, climate change, as well as socio-cultural and economic factors (Berry et al., 2015).

3.3. Local food systems

There is no universal consensus regarding the concept of local food systems (Enthoven & Van den Broeck, 2021; O'Neill 2014; Martinez et al., 2010; Edwards-Jones et al., 2010), but as Nilsson (2009) and Coelho et al. (2018) point out, the term's utilization varies depending on the context. It can even be used interchangeably with "local food", "(re)localization" or "short food supply chains". The main variations arise when defining "local". Different, investigators or modes of government according to their perspectives and regulations have diverse considerations that respond to context when identifying the local level. For example the Government of Canada acknowledges that a food is local when is produced in the province or territory in which it is sold, or commercialized across provincial borders within 50 km of the originating province or territory. For the United States, the scale is larger, it requires approximately 650 km for a product to be labeled as local. In the case of the EU, a food item or a system is local, when produced, processed and retailed within a geographic area of 1 to 100 km radius (Kneafsey et al., 2013).

For Research purposes, local food systems are connected to three aspects of proximity; geographical proximity (e.g. physical locality, distance between food production and consumption), relational proximity (e.g. close relationship between actors within the food system) and proximity in values (e.g. place of origin, traceability, freshness, and quality). Still, geographical proximity constitutes the basis for defining local food systems, while the other dimensions of proximity represent additional features (Eriksen, 2013). However, it can also be defined in terms of social or supply chain characteristics (Martinez et al., 2010).

3.4. Local food systems for improving food security

There are several plausible benefits associated with "localizing the dislocal". Local food systems transcend food production and consumption, recognizing that agriculture, and food systems more broadly, provide multiple, integrated services. (Diekmann, Gray & Thai, 2020) Local food systems seek to be economically viable for farmers and consumers, use ecologically sound production and distribution practices, and enhance social equity and democracy for all members of the community (O'Neill, 2014). As Feenstra (1997, p. 28) stated, 'the development of a local sustainable food system provides not only economic gains for a community but also fosters civic involvement, cooperation, and healthy social relations.

Local food systems involve a range of alternative types of production such as community gardens and urban farms. Taken together, these types of alternative food practice have the potential to make local food available, which in turn support the local food economy, educate people about food and agriculture, as well, as strengthening community (Poulsen et al., 2017). A paramount feature of local food systems is its multi-personal learning component. Evidence signals that local food projects have the quality of co-creating knowledge, a notable example are community gardens, where new gardeners learn through active participation about the biophysical processes occurring in their particular environment (Krasny & Tidball, 2009, p. 2). Additionally, social learning may also take place among a group of gardeners or other stakeholders who come together to address management and policy issues (Diekmann, Gray & Thai, 2020). This also incites community building when, as Feenstra (1997) points out; local food systems facilitate a re-establishing of relationships of trust and accountability between food producers and consumers. To support this notion CPRE (2012) also found evidence that Community Supported Agriculture schemes are characterized by more cohesive communities motivated by several reasons (e.g., environmental concerns, prices, policies).

Another potential quality derived from local food systems, which has been widely studied, is Civic

Engagement. Research by Schmelzkopf (1995) and Ernwein (2014) affirm that some types of localized food production networks lead to political engagement and organization. Community gardens have a history of grassroots political activism against capitalistic forces of development that threaten garden spaces. Associations originating activities related to local food systems, can also empower some gardeners to become more active in their communities (Saldivar-Tanaka and Krasny, 2004; Wakefield et al., 2007). Barron (2017, p. 7) asserts that citizens engaged in local food, whether producers or consumers “cultivate a variety of social and political skills as well as critical perspectives that enable them to participate in promoting food democracy, and motivate the activation of democratic engagement at other scales.”

Reviewing the economic benefits, inquiries in Sweden indicate that economic advantages can emerge from the local food system by selling products directly to the consumer (Nilsson, 2006). Farmers interviewed by Nilsson, stated that a larger percentage of the profit ended up with the farmer/producer instead of retailers. This also lead to more money being generated for the community and staying in the community as members of local systems use their money within the local area, for example, trading with local suppliers. However, the consensus emerging from the Farmers Market study indicated that the income generated from selling directly to the consumer was not the prime motivation force for being involved in the market, the producers admitted that it was more a possibility to market their products and show the consumer “who we are” (Nilsson, 2009). This means that there is more to gain than just economic value. To complement this information, other studies conducted in the global north by the University of Leeds (2018) demonstrated the significant local economic and social value of traditional food retail markets, including through multiplier effects. Local food often supports higher levels of employment relative to sales than national chains.

Additionally, local food is frequently positioned as a core element of regional rural development, as a way to overcome farming crises, and revive lagging rural economies that can restore consumer confidence and thus enhance the rural-urban link (Goodman 2004; Seyfang, 2006).

Furthermore, Edwards-Jones et al. (2008) also considered the nutritional quality foods locally produced compared the non-local. They revealed that for most cases the peculiarities of local food systems, such as shorter supply chains, and briefer times between harvesting and commercialization provide a better dietary value, to which it is possible to add qualitative and maybe subjective elements as “freshness”, “shape”, “color” or odor” that also influences consumer behavior .

Lastly, some qualities of local food systems manifested by the Nyéléni declaration are; that it reduces distance between food providers and consumers; it rejects dumping and inappropriate food aid, and resists dependency on remote and unaccountable corporations that rely on the global food system and market, as well as, traditional supply chains.

3.5. Against LFS

The majority of the known critiques against LFS are directed to the claims that advocates of this model have made asserting its inherent goodness by origin, which considering the non-positivist nature of this thesis, have some truth in them. This means that, LFS in this work are not considered as an infallible “Rosetta Stone” that will enlighten humanity, but an indispensable pathway that has the potential to complement other approaches, and in mutual synergy address and solve current complex problems that afflicts us today.

In this light, some authors such as Kiss et al. (2019) criticized the frequent perception that local is inherently good or more sustainable. Stein and Santini, (2022) manifest that in some occasions local food systems may require more intensive farming to produce enough food to satisfy the local demand. Furthermore, Ritchie and Roser (2020) express, that several elements affect the carbon footprint of a product more than transportation, such as land use, production processes or storage. The carbon footprint of food systems is much more influenced by consumers’ dietary choices than by the

“localness” of the food they buy. When only looking at transportation, “localness” can be a poor guide to determine a product’s carbon foot print (Benis & Ferrão, 2017; Carlsson-Kanyama & González, 2009; Ritchie, 2020; Webb et al., 2013).

Reviewing Pradhan et al. (2015) it is also possible to disclose that in terms of environmental impacts, LFS are not necessarily, more sustainable than systems that operate at larger scales. For instance, local food systems may require intensive farming to produce enough food to satisfy the local demand; on the contrary, agricultural intensification can ensure greater productivity to larger populations.

In the economic dimension, authors such as, Galli and Brunori (2015), Low et al. (2015), Deppermann et al. (2018) or Kneafsey et al. (2013) claim that the Economic value is highly constrained by two factors. (1) Demand for local products may be limited by the number of consumers who can afford local products, or who are willing to do so, which can eventually decrement sales due to low consumer interest. (2) Due to the small scales nature of local systems and the managing of inputs through shorter supply chains, local producer’s profitability may also be constrained depending on how much they can reduce their production costs.

3.6. Action research stance

It is important to disclose that this Thesis is not an action research work. It would have required a longer timeframe involving several iterations to reach this point. Instead, I took an action research stance where I implemented some of the essential components of this approach to my advantage in the research process; such was the case of its learning oriented nature, its aim to solve real world problems, and its participative quality. Furthermore, even if my work did not reach the point of achieving change or taking action, it is key to my stance that my conclusions leave an open path to continue devising ways to tackle the situation deemed as problematic. In addition, to understand better action-research a brief summary of its main features is presented next.

As declared by Serrat, (2017) Action-research is an educational and problem-solving approach that emphasizes learning through action and reflection. It is a dynamic process that involves tackling real-world challenges or projects while simultaneously promoting individual and collective learning. Action research typically involves a small group of individuals who work together to identify and address complex problems or tasks.

The key elements of action research include:

- Real-world Challenges: action research focuses on addressing actual problems or tasks that are relevant to the participants. The challenges can be work-related, organizational, or societal issues that require innovative and practical solutions (Kember, 2000)
- Action: Participants engage in taking concrete actions to address the identified challenge. This can involve planning, implementing, and evaluating different strategies and approaches in a real-world context (Kember, 2000)
- Reflection: Action research places great emphasis on reflection, both individually and as a group. Participants are encouraged to reflect on their actions, experiences, and outcomes, considering what worked well, what did not, and what improvements can be done. Reflection allows for deeper understanding, insight, and learning from the experiences (Marsick & O’Neil, 1999)
- Learning through Feedback and Dialogue: Action research promotes a supportive and constructive learning environment where participants provide feedback to one another. This feedback stimulates critical thinking, encourages new perspectives, and helps individuals refine their approaches.
- Dialogue and discussion within the group enable participants to share insights, challenge assumptions, and co-create knowledge (Serrat, 2017).
- Iterative Process: Action research is an iterative process where participants continuously cycle through action, reflection, feedback, and learning. Each cycle builds on previous experiences,

- Facilitation: A skilled facilitator guides the action learning process, ensuring that participants stay focused on the challenge, encouraging reflection, and fostering effective communication and collaboration within the group. The facilitator helps create a safe and supportive learning environment, facilitates the exchange of ideas, and encourages participants to take ownership of their learning (Marsick & O'Neil, 1999)
- Action research is often used in professional development, leadership training, and organizational learning contexts. It offers a practical and experiential approach that combines problem solving with individual and collective growth. Through action learning, participants not only acquire new knowledge and skills but also develop critical thinking, collaboration, and adaptability, enhancing their ability to address complex challenges effectively (Klopprogge & Sluijs, 2006).

3.7. Soft Systems Methodology (SSM)

SSM are a result of years of an action research program at Lancaster University (Checkland, 2000; Flood, 2000; Ulrich, 1988) that aimed at addressing contemporary science management inability to solve complex real world problems (Stowell, 2016; Checkland, 1981). This means SSM is a problem-oriented approach, which first critical “ingredient” for success is to learn as much as possible from “the situation” and identify the “common problem” and its effects among stakeholders. It is also paramount to express that SSM nurtured by its innate proximity to Action-research and participatory methods, seeks not simply to address a situation or improve the system but to implement change. It as a transformative quality in its purpose. (Checkland, 2000; Stowell, 2016; Ulrich, 1987).

In other words, SSM is an approach for tackling problematical, messy situations of all kinds. It is not only problem oriented, but also an action-oriented process of inquiry into problematic situations, in which users learn their way from finding out about the situation, to taking action to improve it. The learning phase emerges via a structured process in which the situation is explored using a set of models as analytical tools, to inform and structure discussion about a situation and how it might be improved (Checkland and Poulter, 2020)

Peter Checkland, the major exponent of SSM as part of his work, elaborated a Methodology comprised of seven stages, which can be better comprehended in figure 2. Each has its own methods and techniques used to bring forth as much information as possible. A key characteristic of SSM is its flexibility, the stages are not constrained by specific methods, techniques or activities, it is to the practitioner or researcher to learn about the situation, the methodology and adjust to fit the investigation needs (Wals & Sriskandarajah, 2010). In addition, the stages embody a framework into which to place a purposeful activity. The Methodology is composed of two kinds of activities. Stages 1, 2, 5, 6, and 7 are “real-world” activities that necessarily involve people. Stages 3, 4, 4a, 4b concern Systems Thinking and may or not include those in the problem situation (Checkland, 1981).

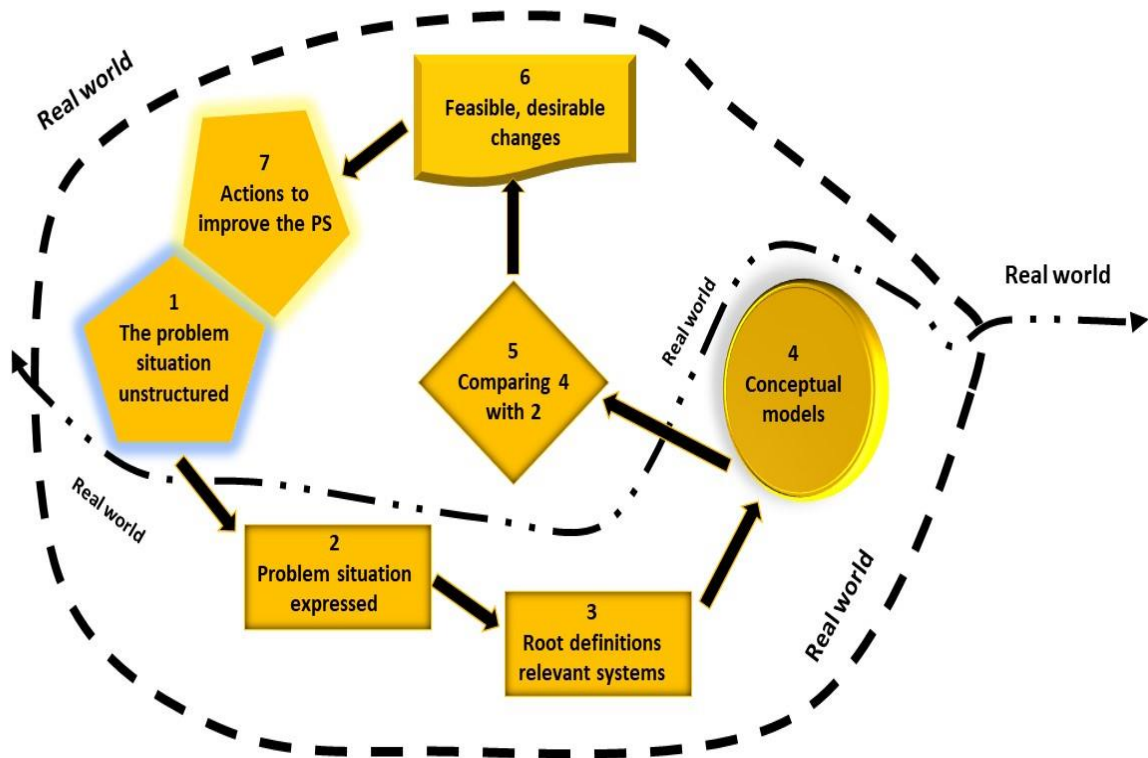


Figure 2. The seven stages of SSM (Built from the original diagram by Peter Checkland, 1981).

The SSM process can also be understood in terms of four main phases: finding out about a real-world problem situation; modelling relevant systems of purposeful activity; comparing perceptions of the real world with conceptual models of relevant systems; and, taking action to improve the situation.

As a framework for my work, I followed through six of the stages declared by Checkland, each one with its own techniques and purpose. The process can be summarized in two main phases; the first that determined if the situation deemed as problematic was indeed problematic, and the second that sought to devise actions for improving the situation. This essential transformative quality of SSM represented an ideal approach to guide this Thesis. Another inherent quality is that, it relies on an interpretivist-based paradigm (Flood, 2000) that can provide a holistic identification of the elements configuring the problematic phenomenon based on the perceptions of stakeholders. In addition, this Methodology informed by systems thinking analyses and synthesizes relations and interconnectedness between actors, the effects that the problem has on them, and then involve them in taking action co-designing potential solutions through participative methods.

3.7.1. SSM for making sense of the methods

As declared above, SSM played a decisive character for my Research process, but not only for specific lapse of time. As a Methodology, it has been the path in, which the methods, approaches and frameworks transited. The stages conceived by Checkland (1984) were a roadmap vital to take.

Trying to makes sense of the techniques and methods throughout the process in a narrative manner, it is possible to say that, for the first stage, the problem situation was unstructured: This phase goes back to the beginnings of the Research Process, when without an in-depth investigation a clear problematic situation manifested itself in everyday life all around. This picture at that moment still blurred, but it was possible to see dimly some elements forming. Elements so immense that felt obvious to note.

There was Climate change, land degradation, then the persistent scars of COVID-19, that brought questions about post-COVID era, or not yet post? Then war, not here, but still not so far. Transversal in time, and in most friendly conversations, notions of having a little less money each month. Thinking digitally, one could say these were bits on an image bitmap, co-habiting in the same context, and if they co-habit, they must be related. During the first stage a profound scrutiny of official documents, and academic literature was conducted as a means so start building a background to the situation, formulate a research problem and questions. Then To determine if these perceived phenomena constituted “a problem”, the second stage incited to investigate further and contrast the preliminary notions. Is here statistical data was obtain, and processed, trying to link, to find causality between one bit of the picture to the next. These two stages showed that the triple threat of Climate change, COVID-19 and Russia’s invasion are causing problematic situations, which are interwoven. At the same time, the opinions of Government officials, IPCC, UN, EU, Researchers and the common citizen were all expressing elements of the same situation. Thus, a common problematic situation was revealed.

Now for the third stage, SSM indicated to identify systems, to pick the system that the Research aimed to improve or change. In this mostly “Thinking in Systems” section, a variety of systems, sub-systems, scales, elements, interrelations, functions and boundaries were put forth into an exploratory mental model. This mental model represented a preliminary comprehension of the situation from a systems view at that particular time. At that moment, the system to improve was determined to be Uppland’s food supply. Other elements such as Uppsala County being the focal system were also set, as well as, some dynamics that helped to understand how to achieve a potential enhancement. Coming to the fourth stage, the mental model grew in complexity; larger systems were delimited, as is the case of the Swedish food system. At this point, it became a matter of scales, incorporating the regional European system and the global food system, each with its own set of elements. Additionally, other functions or sub-systems were considered. This was necessary to establish relations between different elements, also to examine what factors or stressors affected them in positive or negative ways. This was done several times, testing different relations, functions, elements of the sub-systems. A clear example was establishing feedback loops to refine the mental model.

Having examined that the dynamics of the mental model corresponds with each other it was then time for the fifth stage, to contrast the model with reality. At this point is when, the Problem-ranking matrix took place as a way to gain feedback from those involved in the real world situation, and to start co-designing possible solutions to a common problem. For a problem-oriented approach such as SSM, it is vital that those involved in the situation are part of the solution. This activity only served the purpose of mediating between a conglomerate of problems and the people affected by them. The key element here is to acknowledge learning as the fundamental state, which in this case was learning about the situation. Finally, it is paramount to disclose that due to a short time period, this Thesis pauses at the sixth stage by facilitating the conception of a set of actions to achieve change. In other words, this Thesis addressed the situation, but it does not reaches the final seventh stage, which implies transformation. Devising a set of steps to address the problematic situation was accomplished in a focus group that involved the same participants from the Problem-ranking matrix. Stage seven, which is to take action and reach the desired stated would require several cycles or transitions. In some contexts, this also has the name of iterations; this is when the Researcher in a longer timeframe comes back to the situation several times until it improves.

For each of the six stages there is at least one activity, method or technique. It is important to distinguish activities that involved methods for gathering empirical material, which are explained next in the Methods chapter. In the case of the mental models, they were the product of the preliminary literature review, statistical data analysis and interviews, and served as analytical tools to enrich comprehension of the problematic situation.

Table 1. Summarizes the Research process guided by six stages of SSM.

Stage	Objective	Activities and techniques	Purpose
1. The situation unstructured	Obtain preliminary awareness of the context and background information of a situation that appears problematic	Dialogues with family, friends and teachers Literature review	Acknowledge the necessity for further scrutiny and formal structured research
2. The problem expressed	Determine if the situation is indeed problematic	Statistical review Semi-structured interviews to consumers Semi-structured interviews to producers	Methods for gathering empirical material
3. Establishing relations	Gain an integral understanding of the different elements, their relationships and interdependences across scales, as well as setting boundaries for the Research	Mental model	Analytical tool
4. Thinking in Systems	Refine the Mental model in terms interrelations previously conceived	Mental model v.2	Analytical tool
5. Testing the model	Test the Mental Model with the insights gathered from interviewees and questions the accuracy of the real situation	Problem Ranking Matrix	Methods for gathering empirical material
6. Devising action	Facilitate that key actors define a set of steps to take action and improve the situation	Focus group Semi-structured interviews to producers and consumers	Methods for gathering empirical material

3.7.2. Exploratory literature review

For the first stage of SSM, and before implementing any method to gather empirical data, an exploratory literature review was performed. This helped to acquire qualitative insights into the

perceived problematic phenomenon, and select a set of adequate methods, and in consequence delimit the focus of the investigation. Some additional purposes follow Robson et al. (2016:52) where the review helps to:

- Exposed main gaps in knowledge and identify principal areas of dispute and uncertainty.
- Determine general patterns to findings from multiple examples of research in the same area.
- Contrast studies with apparently conflicting findings to help explore explanations for discrepancies.
- Define terminology or establish variations in definitions used by authors
- Pinpoint potential adequate research methodologies and instruments

Table 2. Shows the summary of the literature reviewing process.

Keywords and search strings	Language	Years	Type of publication
<i>“Food security”, “food insecurity”, “food security in the Sweden”, “food security in the Global north”, “Global Food Systems”, “Systems approach to food security”, “Local Food Systems”, “food security and Climate change”, “food security and COVID-19”, “food security and Russia war”, short food supply chains”, “FAO food security”, “World bank food security”, “EU, food security”</i>	English	1945-2023	Government documents (policies, law, bills, reports); Scientific Journals; Book Chapter; Book; Online websites; Master’s Thesis; Doctoral Thesis
<i>“La Vía Campesina”, “Desertificación Chile”, “Soberanía alimentaria”; declaración Nyéléni</i>	Spanish	1994-2023	Scientific Journals; Grassroots movement publications; Online websites
<i>“livsmedelssäkerhet i sverige”, “äppleproduktion”, “regering EU”; mat och covid 19 i Sverige</i>	Swedish	2015-2023	Government documents (policies, law, bills, reports); Online websites; Master’s Thesis; Doctoral Thesis

4. Methods

As expressed in the previous chapter SSM piloted this research, it gave sense to the methods employed for gathering information and created knowledge along the process. At this point, it is important to notice that this chapter exposes the methods used for gathering empirical material, which corresponds to stages; two, five and six of SSM (see figure 2 and table 1). As is usually done in research a exploratory literature review was performed in stage 1, for gaining insight into the problem, as well as, elaborating the introduction, background, and theoretical framework, and does not count as a method for this chapter.

4.1. Research Design

Congruent with a flexibility inherent to SSM that calls for quantitative and qualitative data to be integrated in order to deal with complex real world situations (Mingers & Taylor, 1992), the research design of the thesis is based on Convergent parallel mixed methods. This as explained by Crewswell (2014:44) is a form of mixed methods approach, where the researcher converges or merges quantitative and qualitative data in order to provide a comprehensive analysis of the research problem. In practical terms for this work meant that the qualitative and quantitative methods chosen for gathering empirical data were collected separately without attempting to make sense of what one method meant to another. In addition, the data was also analyzed separately. The converging stage involved comparing and integrating the results from both strands of research, this also implied identify patterns, consistencies, or inconsistencies between the qualitative and quantitative findings. This help to determined areas of convergence or divergence, where the findings from the two strands complement, enrich, or contrast each other. In this sense, the methods moved in convergence to the same objective, which was to obtain a comprehensive understanding of the problem, and in consequence devise plausible solutions. After convergence, came merging the results, where the findings produced an integral interpretation of the research. This required synthesizing the merged results as a whole from both strands, as well as, drawing connections between them. In the case of this thesis, themes, explanations, or emergent discourses were considered.

This design fitted the scope of the research, since the overarching purpose guided by SSM was to understand a complex situation that is composed of several phenomena as first step to devise transformative actions. Likewise, the work also embodies traces of Explanatory Sequential design (Creswell, 2014: 44, 177, 180) when as part of the process, the research involved an initial qualitative investigation to build up a background and a problem formulation. In the case of this investigation, this can be seen in the exploratory document and literature review, which served as a first step to gain insight into the research process and in correspondence select appropriate mixed-methods to start gathering empirical data.

As is the case with this Thesis, the use of mixed-methods becomes relevant when examining Environmental problems because it provides a broader and deeper understanding of the Socio-Natural dimensions that require both quantitative and qualitative approaches that either can provide individually. This is also coherent with the view that each has their respective strengths and weaknesses (Kanazawa, 2017).

4.2. Soft Systems Methodology and mixed-methods

For this thesis, SSM and mixed-methods associated to gain a full and richer comprehension of the complex real-world phenomena being investigated. In the case of SSM, it gave structure and sequence to research process by following the stages explained in the previous chapter. This form of organizing the investigation helped to express an unstructured problematic situation. SSM facilitated a deep exploration of the problem context, and its manifestations by taking into account stakeholders' perspectives both for learning about the situation and for improving it. Regarding mixed-methods, it contributed to a more robust analysis by a process of converging and then merging qualitative and quantitative data. It facilitated the integration of the collected qualitative and quantitative empirical

material to gain full comprehension of the phenomena; such was the convergence of the statistical data with qualitative inputs from the interviews and participatory activities. Overall combining SSM with a mixed-methods design, the research process benefited from the strengths of both approaches, it additionally enhanced the completeness and conclusiveness of the findings in interest of answering the research questions.

4.3. Summary of the Data gathering process

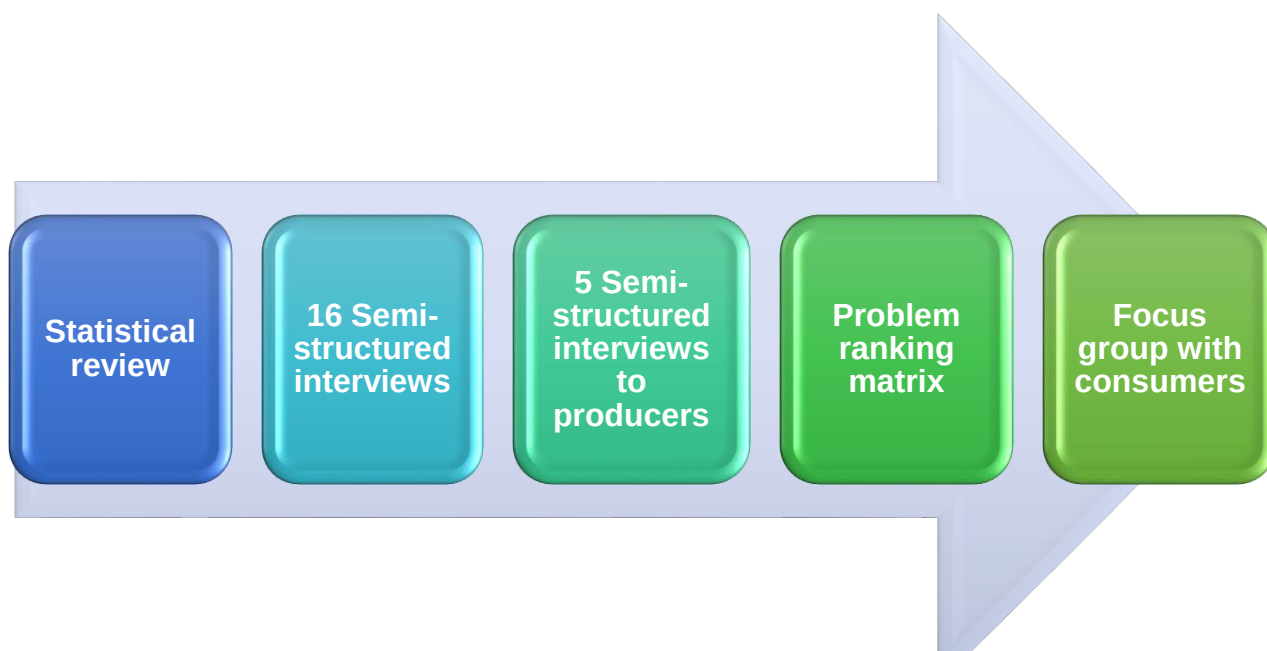


Figure 3. Flow of the data gathering process.

4.4. Statistical review

This method was employed to gain a better understanding of the previous information gathered with the exploratory literature review regarding the problematic situation expressed. More specifically this statistical dimension provided a numerical perspective on how certain aspects (agricultural land, price, costs, imports, exports) that affect food security relate to each other, as well as, revise their behavior in the past years. Additionally, this method served to determine possible vulnerabilities and their causation. For this stage, the main source was Jordbruksverket.se, which is a Swedish website belonging to the Swedish board of agriculture, where a great quantity of unprocessed statistical data and reports are stored. These are the official Government statistics, that researchers and institutions such as the WB, UN or EU use to elaborate their own reports. The results obtained complemented the previous qualitative examination of documents.

The sequence of this method can be compiled as follows:

- 1- Determine a series of statistical areas related to food security in terms of its supply, both in Sweden, as well as, in Uppsala County.
- 2- Within the statistical areas, variables were chosen that related to domestic agricultural production, plus a range of years, which most times it was selected as initial point year the 2000 or after.
- 3- The first two steps produced for every variable a specific datasheet, which was then used to analyze a statistical area.
- 4- Tables summarizing the results from each spreadsheet were elaborated, as well as graphical

tools to facilitate the expression of the numerical values.

- 5- Analyze the statistical data and graphs through inductive association, seeking relations between variable or statistical areas, patterns, and trends (increasing, stagnant, decreasing)

Table 3. Summarizes the process of the statistical review.

Type of statistics	Variables	Years reviewed	Scale	Purpose	Tools for analysis
Total land cultivated by crop (conventional and organic)	Winter wheat, Spring wheat, Rye, Autumn grain, Oats, Rye wheat, Autumn rye wheat, Spring rye wheat, Mixed-grain, Maize, Field beans, Peas, Fall trap, Spring rape, Winter rape, Spring turnip, Oljelin, Straw seed crops (excl. maize) for green fodder, Maize for green fodder, Crops other than straw cereal crops for green fodder, Potatoes,	2009-2022	Uppsala County	Determine shrinking patterns by statistical inductive reasoning in the individual crops cultivated in the County	1-Table showing values for each crop and presenting it current state 2-Bar charts
Total agricultural land	cultivated, arable, pasture, under conversion, both conventional and organic	2009-2022	Uppsala County Sweden	Establish trends in the total agricultural land by using statistical inductive reasoning	Line graphs
Total harvest and yield by crop (conventional and organic)	Winter wheat, Spring wheat, Rye, Autumn grain, Oats, Rye wheat, Autumn rye wheat, Spring rye wheat, Mixed-grain, Maize, Field beans, Peas, Fall trap, Spring rape, Winter rape, Spring turnip, Oljelin, Straw seed crops (excl. maize) for green fodder, Maize for green fodder, Crops other than straw cereal crops for green fodder, Potatoes	2009-2022	Uppsala County	Determine shrinking patterns by statistical inductive reasoning in the individual crops cultivated in the County	1-Table showing values for each crop and presenting it current state 2-Bar charts

Type of statistics	Variables	Years reviewed	Scale	Purpose	Tools for analysis
Imports vs Exports	Fruits and vegetables	1999-2022	Uppsala County Sweden	Comparison for establishing dependency on food imports	
Number of companies with organic and conventional farming	Organic and conventional farming	2003-2021	Uppland Sweden	Determine shrinking patterns by inductive reasoning in the individual crops cultivated in the County, and then associate its results to other variables, such as total agricultural land, and land prices.	Bar charts
Total animal production (conventional and organic)	Cows, calves, poultry, bulls, rams, heifs, ewes, lambs, boars, sows, pigs, piglets	2015-2022	Uppsala County	Determine shrinking patterns by statistical inductive reasoning in the individual crops cultivated in the County	1-Table showing values for each crop and presenting it current state 2-Bar charts
Agricultural Land prices	Organic and conventional farming; rented or owned	1990-2022	Uppsala County Sweden	Determine trends by statistical inductive reasoning, and then associate its results to other variables, such as total land, and land prices, and number of	Line charts

Type of statistics	Variables	Years reviewed	Scale	Purpose	Tools for analysis
				agricultural companies	
Production costs vs production index vs consumers costs	Reviewed agricultural report from year 2021 and 2020 at jordbruksverket.se/	2020-2021	Uppsala County Sweden	Associate by using statistical inductive reasoning the values to other trends, as land prices, number agricultural companies and total agricultural land	
Fruit production	Apple, cherry, plum, pear	1999-2022	Uppsala County Sweden	Determine trends and historical behavior of these specific products in the County and Sweden	1-Table showing values for each crop and presenting it current state 2-Bar charts

4.5. Semi-structured interviews

Semi-structured interviews were used for gaining perspective on key actors that are part of the Research context, as explained by various authors (Kanazawa, 2017: 318; Franklin & Blyton 2011).

For the design of the interview guide the research considered Stephen (2007, p205) who manifests that in semi-structured interviews, there are generally no set or fixed questions. Instead, the researcher, following a review of relevant literature, generates a list of topics or themes that are to be discussed with the interviewee.

In addition, in congruence with Beardsworth and Keil (1992, p261) the topics evolved as the interviews progressed. The questions formulated were at first introductory, and fit the purpose of situating myself as interviewer, and the interviewee into the context. Then, continued by the more 'directive', probing questions (Hammersley & Atkinson, 1989, p113) that sought to understand a particular issue or challenge the interviewees to elaborate on their opinions. To complement these previously generated questions, the improvisation of follow-up questions was highly substantial in terms of constructing meaning.

The interview guide had two versions based on the same approaches but slightly different for producers where there are questions that examine directly their perspectives as business owners based

on their experience. Both guides can be found under the Appendices chapter.

4.6. Participants recruitment

For recruiting the local producers, a contact list was made using the website <https://www.gardsnara.se/uppsala>. Then an email was sent to 12 different producers related to fruits and vegetables. The ones that responded are the producers that were interviewed for a total of five in Uppsala County. Interviews took place in the respective farms premises. This implied taking walks along the complex (hangars, depots, plantations, product-lines) or just talking in the office of the business.

Regarding the interviews to consumers, there were a few parameters implemented. First all activities were conducted inside food stores in different places in Uppland. (1) ICA Supermarket in Storveta, (2) COOP in Tierp, (3) Willys in Gottsunda. At the same time, some diversity was attempted in relation to age, gender, origins and having a family.

Table 4. Condenses the participant recruitment.

ID	Place	she/her-he/him	Age	Characteristic
CI1	ICA Storveta	she/her	55	Head of the family
CI2	ICA Storveta	she/her	36	Single mother
CI3	ICA Storveta	he/him	25	Long-term immigrant worker
CI4	ICA Storveta	she/her	23	Student
CI5	COOP in Tierp	she/her	62	Retired
CI6	COOP in Tierp	he/him	39	Business owner
CI7	COOP in Tierp	she/her	38	Head of the family
CI8	COOP in Tierp	he/him	53	Has Family
CI9	COOP in Tierp	he/him	47	Unemployed
CI10	Willys in Gottsunda	he/him	50	Policeman
CI11	Willys in Gottsunda	she/her	24	Student
CI12	Willys in Gottsunda	she/her	26	Student/immigrant
CI13	Willys in Gottsunda	he/him	66	Retired
CI14	Willys in Gottsunda	he/him	38	In charge of grocery shopping

ID	Place	she/her-he/him	Age	Characteristic
CI15	Willys in Gottsunda	he/him	29	Large family
CI16	Willys in Gottsunda	she/her	24	Single mother/immigrant

4.7. Ethical reflection

My only ethical concern was a sense of responsibility to announce my presence and purpose to one of the clerks of the store in case somebody found me strange or annoying. A second and maybe more important step for me was to, carefully (also slowly) and politely declared my intentions to the people I approached and wait for a clear informed consent from their part. I tried to follow the explanation used by Davies (2008, p. 55) referencing Homan (1991: 71-4) where two elements should be considered. “First, informing participants of the nature and likely consequences of their participation in the research in a way that is comprehensible to them; and, secondly, obtaining consent that is based on their understanding of this explanation and free of any coercion or undue influence”.

4.8. Participatory Activities

These activities represent the corner stone of the Action-research (MacDonald, 2012; Danley & Ellison, 1999) quality that this work seeks. These activities were conceived to involve the people previously interviewed (both producers and consumers), but it was not possible to accomplish. Instead, the participants came from my local Neighborhood in Sävja, Uppsala. In this case, it was not possible to choose a great variety of backgrounds. I contacted potential participants through the Neighbors association, and based on their willingness to engage, six people were selected. The activities were meant as an exercise for creating spaces for dialogue within the community and future deliberation, where issues of food security, as well as, increasing local cultivation were discussed. Conversations were recorded with informed consent and transcribed in order to be analyzed.

4.8.1 Problem-ranking matrix

This particular technique is mostly used in participatory rural appraisal as a way to prioritize problems or needs in a community. Ranking and scoring exercises are used to analyze, ponder and present information. This exercise proved a powerful mode of representing quite complex information in a way that is clear to both the researcher(s) and the local participants. This also allows for sets of data to be compared, discussed, adjusted and looked at from different points of view in a way that is very difficult in a simple discussion with no visual reference (Sontakki et al., 2019).

The sequence of the activity was as follows:

1. Present the topic or problem and Select the sub-items with the participants
2. Participants choose among the sub-items a set of problems or preferences to be prioritized
3. Facilitate the participants to list the criteria.
4. Drawing of a Matrix using the items declared by the participants
5. Each participant has a page with the same matrix and they rank the options
6. Calculation of the results from each participant’s matrix.

4.8.2 Focus group

To complement the problem ranking Matrix and motivated by (Davies, 2008, p.106), who declares that in policy-oriented research: focus groups are becoming popular, when the Investigator directly tries to introduce the topic into the structure of the discussions. In the specific case of this thesis, the

implementation of the focus group plays a key role providing the participatory quality that is sought. Some of the benefits of engaging in a focus group for this research are:

1. Provides a space for those affected by the problematic situation, where they can actively participate in the research process and devise potential solutions. This helps to bring together individuals with diverse perspectives, experiences, and expertise. Additionally focus groups allow participants to contribute their experiences, insights, and concerns related to the research topic. This engagement fosters a sense of ownership and empowerment among participants, as they become active and involved, rather than passive subjects of the research (Bagnoli and Clark, 2010)
2. Generate knowledge that is relevant and useful for stakeholders. In this sense a focus group, facilitate the co-creation of knowledge by encouraging dialogue, discussion, and exchange of ideas among participants, which enriches the research findings and ensures their practical relevance. This collaborative process allows for the integration of different perspectives, resulting in a more comprehensive understanding of the research topic, which also gives elements for answering the research questions (Bagnoli and Clark, 2010).
3. Identification of priorities that can occur by engaging in discussions and sharing their concerns, participants can shape future the research agenda, helping researchers to focus on issues that matter most to the stakeholders. This participatory approach ensures that the research is responsive to the real-world challenges and needs of the community or group being studied (Morgan and Spanish, 1984).
4. Potential to bridge gaps between research and practice by engaging with participants in discussions. In this way, is possible to gain input into the practical implications of the research findings and how they can be applied in real-world settings. This collaboration facilitates the translation of research outcomes into future actionable recommendations, policies, or interventions that address the needs of stakeholders and this improve the situation (Lane et al., 2001).
5. Looking beyond the scope of this thesis, a focus group in participatory research enable stakeholders to participate in decision-making processes. By actively involving stakeholders in the discussions, there is the possibility to gather their input on key decisions related to the research design, data collection methods, and analysis. This inclusive approach ensures that diverse perspectives are considered and that the research process reflects the needs and priorities of the stakeholders. It enhances the transparency, legitimacy, and acceptance of the research outcomes. (Davies, 2008, , p.107)

The Questions for the focus group were designed based on the previous inquiries during the interviews, as well as, considering the basic themes of the problematic situation (worldwide land degradation, COVID-19, Russian invasion, food security, vulnerability, paths for taking action, localizing food production). The same people from the problem-ranking matrix participated in this activity, this means that six persons engaged in the discussion. The activity lasted around one hour and half. The main considerations taken for the focus group, were first the results from the problem-ranking matrix, and in a second phase the questions that were previously elaborated for fostering the discussion.

The questions guide for this exercise are in the Appendices A.

4.9. Methods of Analysis

4.9.1. Analysis of the interviews

For the analysis of the interviews, this work implemented one the versions of Thematic Analysis. This particular variant comes from Khokhar et al. (2020). To reinforce the examination this process will also utilize thematic guidelines proposed by Gupta, Shaheen and Reddy, (2019). Synthetizing the procedures from these authors and adapting it to the purpose of this work, the process can be summarize as follows:

1. Familiarizing with the data by reading it multiple times.
2. Initial coding to identify meaningful units of data and label them with descriptive codes, for then selecting a group of similar codes to form initial themes.
3. First theme refinement, ensuring they accurately represent the content, and context of the data.
4. Organize the themes in relation to the initial topics proposed by the interview guide.
5. Contrast themes in relation to the initial topics proposed by the interview guide by examining examples or quotations from the original data.
6. Second theme refinement, ensuring they are distinct, coherent, and representative of the data.
7. Define, and revise name of the themes by assigning clear and descriptive names that capture their essence.
8. Interpretation, that in terms of this specific thesis is to analyze and synthesize the themes in relation to the research questions or objectives.

4.9.2. Analysis of the focus group

The initial plan for analyzing the focus group was to use the exact process as the interviews. However, during the activity the participants proposed an alternate manner to analyze the focus group. They requested a participative scrutiny, which implied a slightly different approach than the traditional thematic or content analysis. This in practical terms, means that the procedure was followed but in collaborative way. Additionally, step one was suppressed, the participants started by selecting a wide range of themes that originated from the questions and discussions. After having this first round of themes, these were associated in terms of their content. For example, home gardening, local food, and investments in food genetic research were associated together. Afterwards this associations were synthesized in a brief message, this was called giving meaning to the links between themes. The messages were associated in order to form a new message. At the end four messages composed of the previous themes remained. From these, a final message was constructed carrying the essential themes and content, at this point the collaborative dynamic ended. This process was then expressed in the elaboration of a mind map, where it is easy to trace back and forth or vice-versa the themes and their sense making into small meaningful messages.

4.9.3. Analyzing the statistics

For processing the quantitative data, basic (without computing tools and just inductive and deductive reasoning) statistic associational analysis was employed. This analysis technique has significant connections with functional thinking and other areas of mathematics, such as probability and proportional reasoning (Batanero, Godino & Estepa, 1998). For this thesis associational statistics serve a tool helped to make predictions and find causation, it was also implemented to establish relationships among multiple statistical topic areas, as well as, variables. In addition, it was utilized to determine logic inferences and predictions regarding the statistical data and graphics produced. This involved collecting, organizing and analyzing statistics based on established variables declared for this particular research method, and then, identify decreasing and increasing patterns for establishing trends for both the Swedish and Uppsala county context. The main goal for performing association in this research was to find causal patterns or trends, which help us to understand the situation. It is important to point out that, for this work; association does not necessarily imply a causal relationship.

5. Results

In this chapter, the most relevant findings are presented separately. The order follows the stages of SSM where each of them had a particular technique or method for producing results that informed the research questions. This also means that the mental models are disclosed as a purposeful product of stages three and four even if they were not used for gathering empirical material but as analytical tools. Other findings are available for review in the Annex Chapter.

5.1. Statistical review

This method corresponds to the second stage of SSM, and is the first used to gather empirical material. It helped determine with statistical data, certain patterns, and trends in variables that provide a quantitative approach for understanding if the situation explored by my thesis was indeed problematic. In addition, it provide preliminary insights into a range of plausible vulnerabilities that need to be put in contrast with the results from other methods. The findings exposed in this chapter come from processing statistics coming from The Swedish Agency for Agriculture's statistical database. Most of the results expressed were created specifically for my work using their online tool that facilitates choosing among, statistical topic areas, specific variables within those areas, and a range of years. This process had for an outcome a great quantity of spreadsheets, which then facilitated the processing of the information via graphs and inductive associative analysis. In some cases, the data comes from reports elaborated by the Swedish agency that have processed values. A detailed explanation of the process can be found in the Methods chapter. Additional graphs and tables are in the Annex. For this chapter the most relevant and interconnected results are presented.

5.1.1. Agricultural land is decreasing both in Sweden and Uppsala County

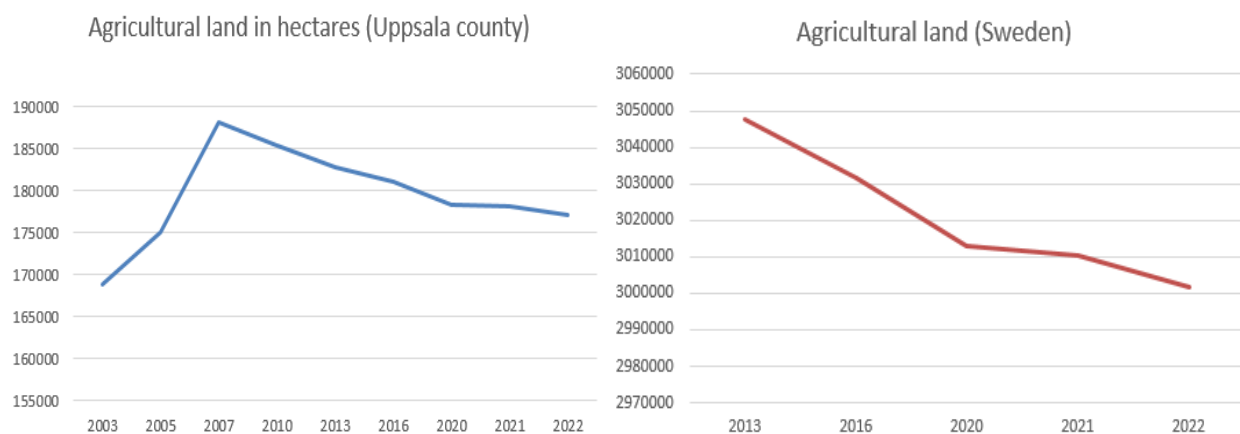


Figure 4 and Figure 5. Agricultural land in Uppsala and Sweden

In **Figure 4**, it is possible to appreciate that Agricultural land in Uppsala County has decreased 5.89% since 2007, and In **Figure 5**, it is possible that in Sweden, has decreased 1.5% since 2013.

These results were produced combining the statistical topic areas of total pastures, total cultivated agricultural land, areas converted for agricultural purposes and under conversion. At the same time, a range of years was chosen. This procedure was applied to Sweden and Uppsala County, and the result is the total Agricultural land.

5.1.2. Prices

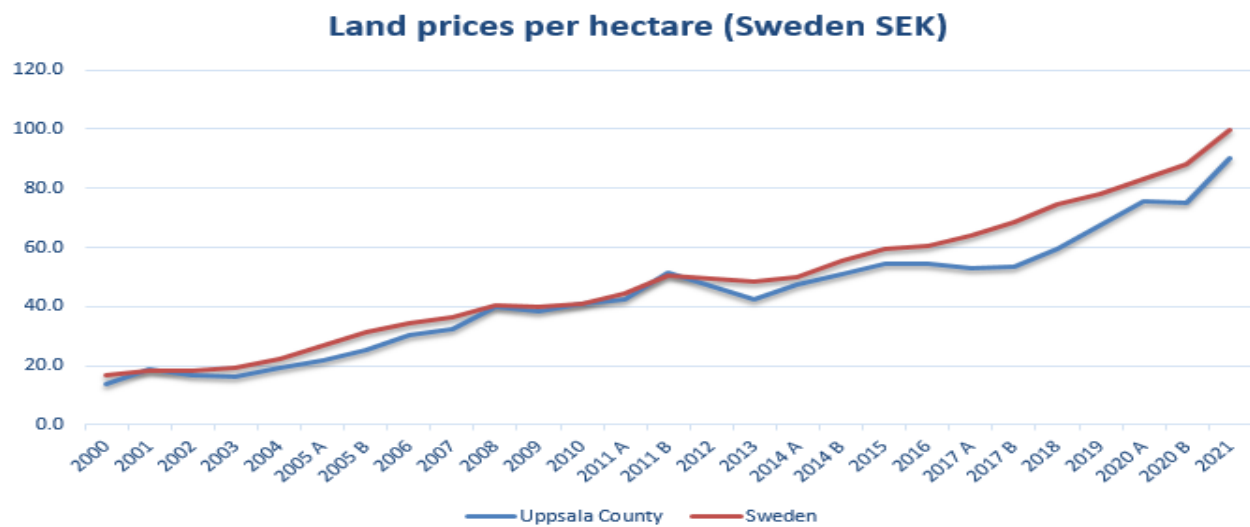


Figure 6. Data from The Swedish Agency for Agriculture's statistical database reveal that since 2000 agricultural land prices in Uppland have experience a 549.64% increase and in Sweden a 494.643% increase. This result was produced selecting statistical topic area of agricultural land prices and Sweden and Uppsala County in separate datasets and then in a new spreadsheet combine both values for producing the results expressed in the line chart.

5.1.3. Limited capacity to grow

In Sweden as a country the number of micro agricultural companies (0 to 2 hectares) has grown over the last fifteen years, but small and medium business show a decreasing pattern.

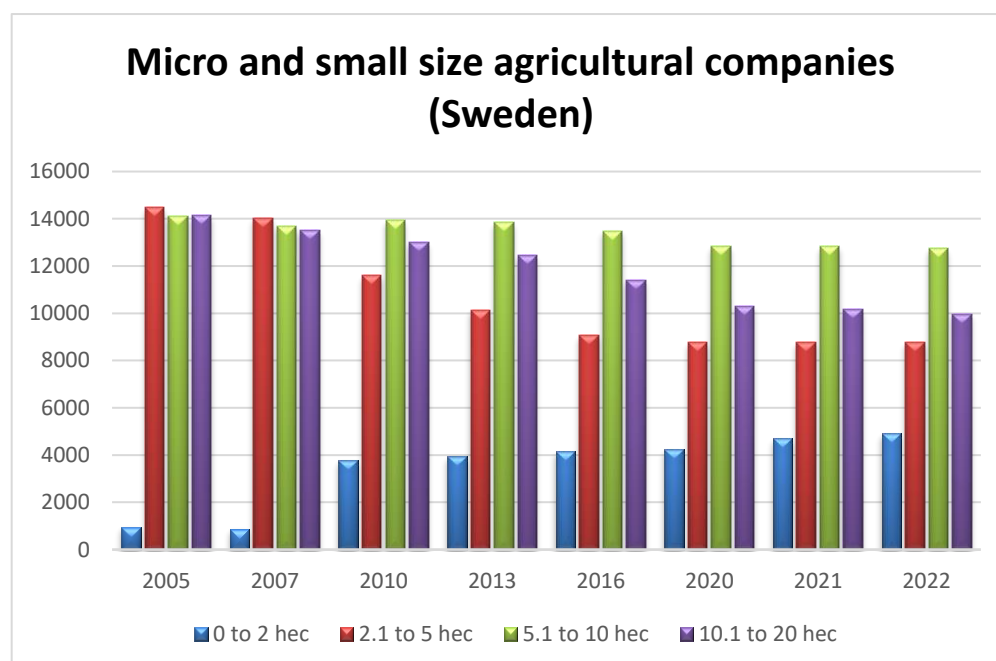


Figure 7. Behavior of number of micro and small size agricultural companies in Sweden. These results expressed in the graph above were produced selecting statistical topic area of agricultural companies (both organic and conventional) in Sweden in separate datasets. Also, as variables small to medium farmers were selected (Jordbruksverket considers 0 to 2 hectares Micro agricultural companies and 2.1 to 20 hectares Small farmers.) Then in a new spreadsheet combined both values

for producing the results expressed in the bar chart.

For Uppsala County in terms of micro agricultural companies, the pattern is similar to Sweden as a country. Beginning in 2010, there is a small but steady increase in number over the years. Concerning companies with 2.1 to 5 hectares, they experienced decline around the year 2007, since then the number has fluctuated with small increases and decreases, but it has not gone back to as many as they were in 2005. Contrary, agricultural business from 5.1 to 10 hectares have grown in numbers. Lastly, medium 10.1 to 20 hectares companies have shown a small reduction of 11 since 2013.

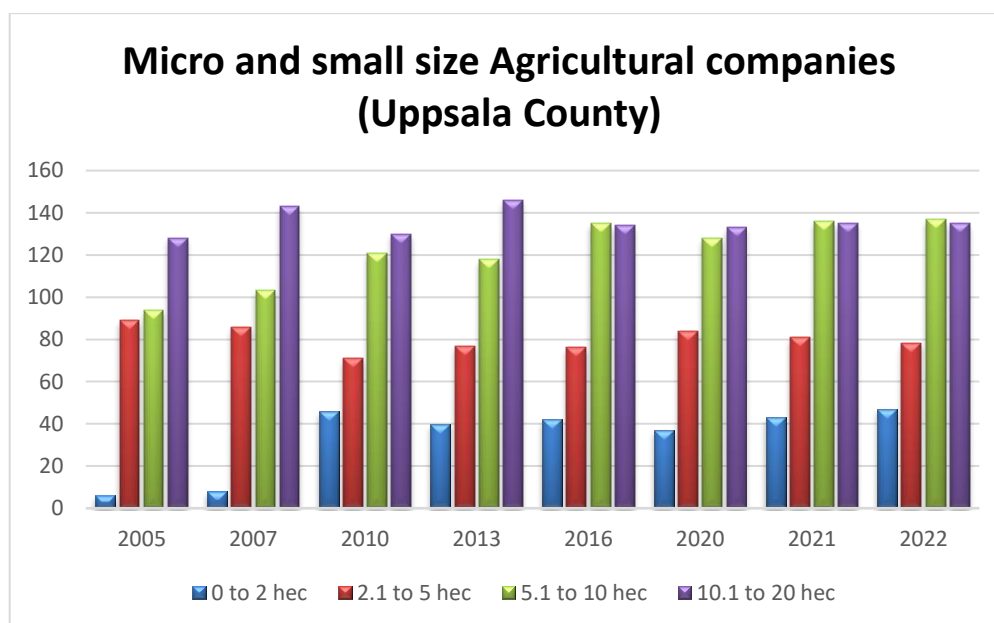


Figure 8. Behavior of number of micro and small size agricultural companies in Uppsala County. The results expressed in the graphs above were produced selecting the statistical topic area of agricultural companies (both organic and conventional) in Uppsala County in different datasets. Also as variables, small to medium farmers were selected (Jordbruksverket considers 0 to 2 hectares Micro agricultural companies and 2.1 to 20 hectares Small farmers.) Then in a new spreadsheet combined both values for producing the results expressed in the bar chart.

5.1.4. Organic Agriculture statistics review

In relation to the notions of food security and local food systems, it was important to examine some elements related to organic production both in Uppsala County and in Sweden. In the section the most relevant findings are presented, however in Annex C are a great quantity of statistical data related to individual crops, animals, cultivated area, and yield. It can be noted that in **Figure 12**, that Uppsala County companies labelled as organic are declining since the year 2019 when there were as much as 279 compared to the 247 recorded by the end of 2022 (-11.5%). Examining **Figure 13** the behavior for Sweden is almost identical. The highest number was in 2019 with 5,658 companies, which has decreased to 5,018 in the year 2022. (-11.3%).

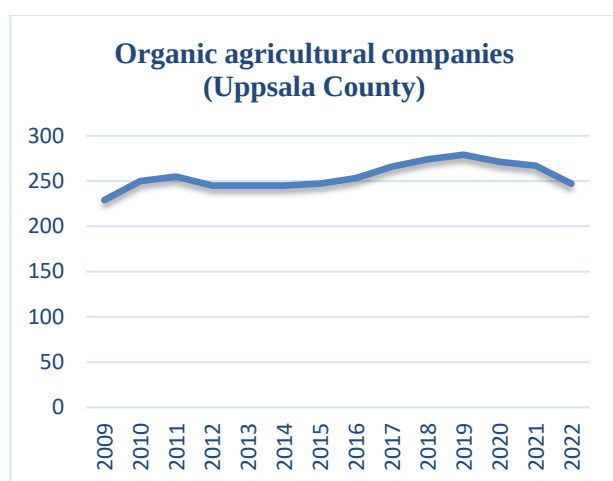


Figure 9.

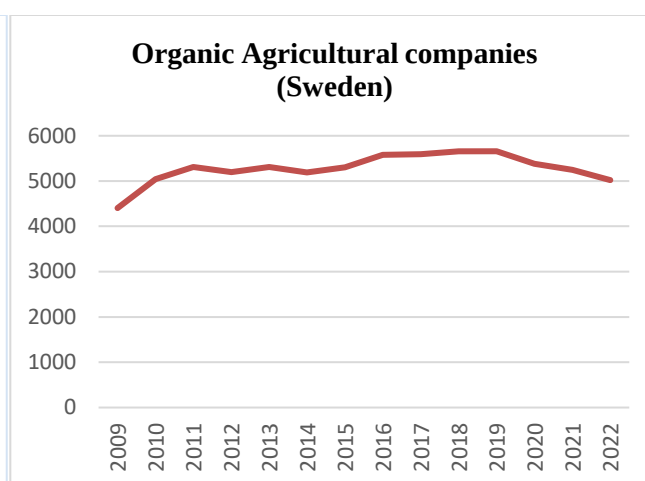


Figure 10.

The results expressed in the graphs above were produced selecting the statistical topic area of organic agricultural companies, both in Sweden and Uppsala County (in separate datasets). For these statistics, it was not possible to choose the size of agricultural business as was done in the previous for the processing of the previous variables presented. Afterwards in a new spreadsheet the Uppsala County, and Sweden values were combined for producing the results expressed in the bar chart.

In 2022, 606,900 hectares of agricultural land were converted or under conversion to organic production. This corresponds to 20% of the total area of agricultural land in Sweden. Compared to 2021, the area decreased by 3,900 hectares or barely 1%. The area of agricultural land that was converted to organic production increased in 2021 by 3,800 hectares or barely 1% compared to 2020 and amounted to 570,400 hectares. The converted area's share of the total agricultural land in Sweden amounted to almost 19% in 2021.

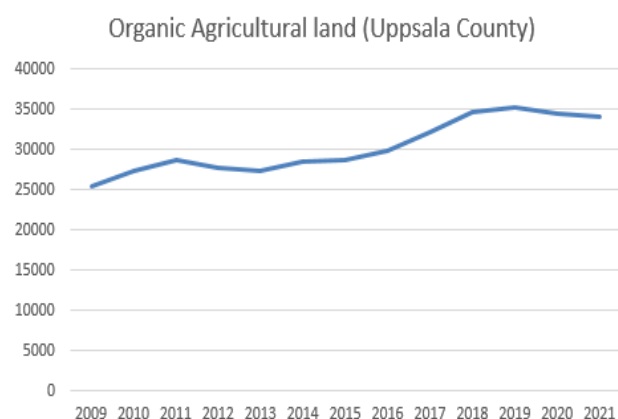


Figure 11. Organic Agricultural land has decreased 2.6% since 2018 in Uppsala County.

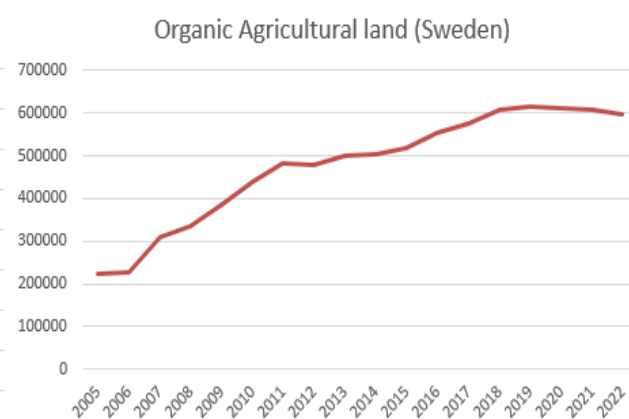


Figure 12. Organic Agricultural land In Sweden has decreased 2.7%

These results were produced combining the statistical topic area of organic agricultural pastures, organic agricultural cultivated land, organic agricultural land under conversion and converted. (Both for Sweden and Uppsala County in separate datasets). This combination resulted in the total organic agricultural land for the country, as well as, County. Afterwards in a new spreadsheet the Uppsala County and Sweden values were combined for producing the results expressed in the line chart.

5.1.4.1. Panorama of Uppsala organically cultivated land

In 2021, the total organically cultivated agricultural area in Uppsala shows a similar dynamic to Sweden's. It has decreased by 0.9%, which is close to the 1% reduction for the whole country. The total organic area includes both land that has been converted and is being converted to organic production as well as the categories of arable land and pastures.

It is important to consider that even if the total area is decreasing it is possible to note that converted land has increased 1.5% in relation to 2020. In the specific cases of pastures and arable land, it increased by 1.9% and 1.4% respectively. To contrast these increments and reveal the reason for the contraction in the total agricultural area is that the current land under conversion has declined in comparison to previous years. Compared to 2020 it decreased by 28% compared to 2020. Moreover, in the case of arable land, it diminished by 35%. The only growth is in pasture areas being converted with 25%. Another takeaway is that Pastures is the only category that in both lands converted, and under conversion to organic has expanded which results in a total of 3.4% increase compared to the year 2020. Regarding Uppsala food products it is complex to synthesize what is taking effect. Taking a general vista, some of the results that are possible to perceive is that while there is some growth for some crops, such is the case of winter and spring Wheat, autumn currants, and Peas, most crops are experiencing a decline in yield, crop area, total harvested tons and companies. For organically kept animals the situation in comparison to the year 2020 is slightly more balanced, where the amount of total cattle contracted by 0.4%, the number of total pigs also decreased by 22%, but the number of sheep and lambs augmented by 7.5% approximately. It is essential to mention that even if the statistics have been taken from the Swedish board of agriculture, there are still gaps in some years for several products. On other occasions, there is no data even if there was production. A critical example to mention was the unavailability of data related to the total tons of fruits and vegetables imported. The only information was the total import in monetary terms.

5.1.5. Consumption of Vegetables and Fruits

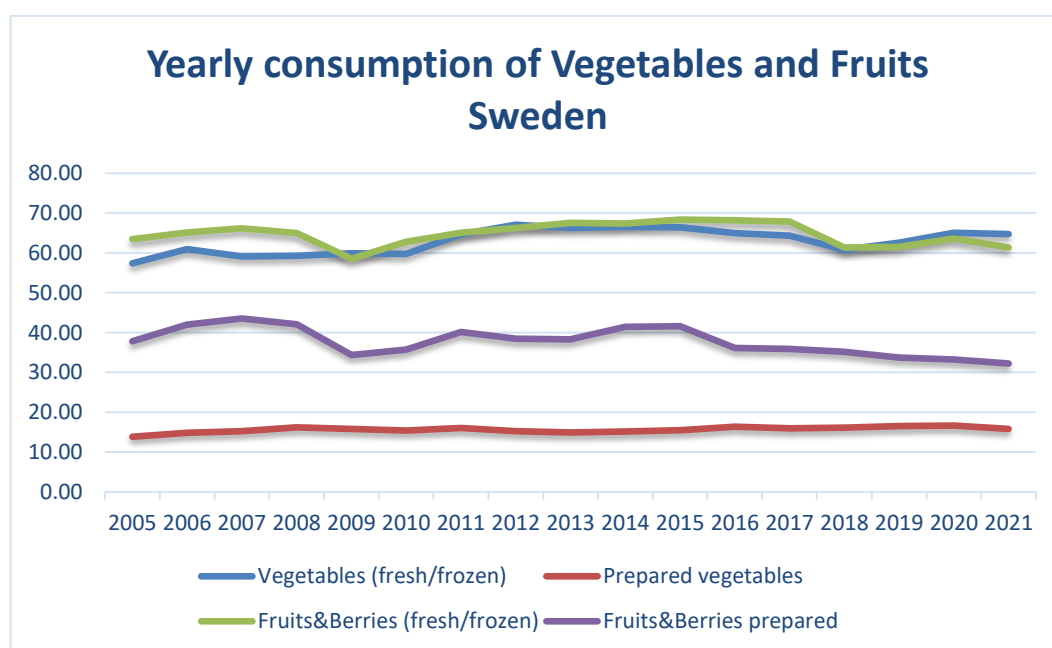


Figure 13. Reveals a quasi-stagnant pattern in yearly consumption of fruits and vegetables. In the case of fresh/frozen and prepared vegetables, there has been a persistent slight increase. The results expressed in the graphic above were produced selecting the statistical topic area of consumption; these values were only available at country level. It was not possible to examine the Uppsala County variable. Within the consumption statistics, vegetables (fresh/frozen and prepared), as well as, fruits and berries (fresh/frozen and prepared) were chosen. Each of this variables was produced in an

individual dataset. Afterwards in a new spreadsheet the four variables were combined for producing the results expressed in the line chart.

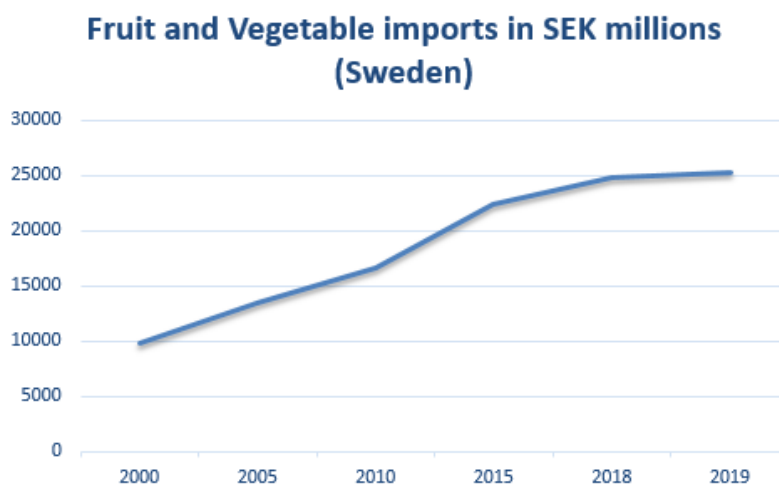


Figure 14. Exposes a rise in imports in monetary terms, coming almost to a stagnant state. Since the year 2018, the increment has slowed down to a 1 % yearly increase in million SEK. These results were produced by selecting the statistical topic area of imports (in monetary terms) for Sweden and then selecting the variable of fruit and vegetable. This assembled a spreadsheet that was used to build the results expressed in the line graph.

5.2. Results of Semi structured interviews to consumers and producers

This section presents the results of the qualitative, semi-structured interviews conducted with sixteen consumers in Uppsala County, and five local organic producers from Uppsala County. This method corresponds to the second stage of SSM, where the problem situation is expressed. The interviews aimed to gather individual perceptions from the concerned actors on a common situation, as a participative way to define what the problem is, as well as its different manifestations and elements.

As expressed in the method chapter, the interviews to consumers were performed inside the buildings of different food store throughout Uppsala County.

To summarize the findings, the content is grouped in Themes and sub-themes. This is also coherent with the interview guide (Appendix) that was designed to extract information from different subjects that compose the main topics examined through this work. In addition, the results are presented in four different tables that relate to the two main phases of SSM identified in this work. The first understand a common problematic situation, and the second to devise plausible steps to improve

Table 5. Situating the problem with consumers

<i>Aim</i>	<i>Themes</i>	<i>Sub-theme 1</i>	<i>Sub-theme2</i>	<i>Sub-theme 3</i>	<i>Examples</i>
Situating the problem	Climate Change or Land degradation, COVID-19, Russian invasion of Ukraine are seen as news	Increment in the cost of living	Stopped buying/reduced some products	Media not helping to understand the situation	CI10: “On TV they say that everything is going to get more expensive due to the war, but I do not see how this affects all food. I have been forced to buy less and less each time.” CI16: “I wonder if coffee prices will continue to rise, I wonder if I will have drink just 1 cup a day, I wonder if I can really by organic food, I wonder if it will get worst.”
Climate change, COVID-19, Russian Invasion	Situation without precedent	Pandemic and War make more damage than Climate change	Security	Insecurity	CI2: “I do not remember a time that had catastrophes as these, what they call climate change, has been there for a while, but in Sweden, I think we don’t feel it that much in the economy, but with Russia invading and the aftermath of the pandemic, it surely feels bad. I particularly do not feel insecure, but wouldn’t be so sure about other people.” CI6 “It seems the Swedish Government is not doing enough to secure resources, make things better, I don’t think that there is a big problem right now, but it comes in waves, we are just feeling the start
Difficulties	Consumers are experiencing a hard time	Local producers experiencing a hard time	Producers seek benefit	Friends or family also experiencing effects of the increment prices	CI14 “It has always been that way, I guess, producers only look for the profit, we people are left to deal with it, everyone is dealing with it, but most people do no say. But talk to your close friend and family, and they will tell you.”

<i>Aim</i>	<i>Themes</i>	<i>Sub-theme 1</i>	<i>Sub-theme2</i>	<i>Sub-theme 3</i>	<i>Examples</i>
					CI8 <i>“It is crazy, right? Many things have doubled; I wonder how this will affect us buyers in the long term. I bet many producers are also having trouble with the situation, I have heard fertilizer prices have gone up.”</i>
Food Security	Sweden is food secure	Crisis are testing Sweden’s capability			C01 <i>“I wouldn’t say that Sweden has food insecurity, I don’t think it will be so, unless something really really really bad happens.”</i> C03 <i>“These crises that you say are putting the country’s capabilities into the test. The Governmental agencies just have to adapt as the situation unfolds, I do not think that the situation worsen.”</i>
Vulnerabilities	Globalized world has its drawbacks	Times of peace have soften policies	European Market		CI5 <i>“Well I think the globalized world is a solution to everything, you that study this, probably know that sometimes it creates disparities, even at regional levels, the European market has not been, let’s say fair to many producers, not the agricultural sector.”</i> CI9 <i>“I remember about 40 years ago that Sweden was sort of self-sufficient in several ways. That was because of the cold war, since that, without the threat of war, measures have lighten, or maybe the European market is another sort of theater of war, you know, economic war.”</i>

Table 6. Devising solutions with consumers

Aim	Themes	Sub-theme 1	Sub-theme2	Sub-theme 3	Examples
Ways to improve	Dietary choices	Citizen manifestation	New measures	Subsidies to farmers, reduce their costs, reduce prices	CI7 <i>“For me what we eat also plays an important role, we could eat cheaper stuff that is also healthy, but we choose otherwise, I think the media if influencing this.”</i> CI11 <i>“One aspect that I consider important is to have real capacity to manifest ourselves and be heard. We should demand new measures to deal with these issues. Something that may help is to provide more subsidies to local farmers especially.”</i>
Civil involvement	Real Civil involvement	Structured Organization	Policy		CI4 <i>“It seems that Swedes have lost a bit of their capacity to shape policy. If I remember well some decades ago there was more organized citizens capable of real involvement.”</i> CI15 <i>“I feel that to some degree the current policies are not representing civil society, we have communal leaders, but it doesn’t go all the way to the bottom.”</i>
Authorities	Not to blame a specific Government but authorities in general	Authorities need a push from the civil society	Some Authorities are not representative	Authorities are not doing enough	CI13 <i>“I wouldn’t blame a particular party or Government, they all have some responsibility in this, they could be doing more to get things better. I would gladly become a representative if they pay me enough (laugh).”</i> CI09 <i>“It may be not the most usual way, there are times that our own Government needs a bit of shaking or a push from the common people, so they know that what</i>

					<i>they are doing is not enough or not helping our situation”.</i>
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Table 6. Situating the problem with local producers (FI= Farmer interviewee)

<i>Aim</i>	<i>Themes</i>	<i>Sub-theme 1</i>	<i>Sub-theme2</i>	<i>Sub-theme 3</i>	<i>Examples</i>
Situating the problem	Climate Change or Land degradation, COVID-19, Russian invasion of Ukraine are seen as news	Increment in the costs of production	Less income	Maintain prices or increase	FI5 “The current situation is getting worst, we are each time gaining less and less, and we cannot increase prices, we have to have them at least equal to other products that come from outside Sweden. In our farm we lost many opportunities because of COVID and we were just getting better our finances when the war in Ukraine started, another shock!” FI2 “I agree that this triple treat that you mention is doing bad for us organic growers, I am sure that there are many like me that want to increase price in the markets, but then nobody will buy, so the only alternative is to earn less and sell at lower at equal price than farmers outside the country.”
Climate change, COVID-19, Russian Invasion	Pandemic and War make more damage than Climate change	Security	The future of the situation		FI05 “Well, I have not felt here the effects of climate change, at least not directly, but the Pandemic and the war did harm to us. If the situation continues like this or gets worse, I doubt that there is going to be a decent future for the very small cultivators.” FI01 “Me being a tiny or micro farmer, I have a small margin of profit that is threaten by how things are developing now. I have not 100% security that my future or my family’s future is secured.”

<i>Aim</i>	<i>Themes</i>	<i>Sub-theme 1</i>	<i>Sub-theme2</i>	<i>Sub-theme 3</i>	<i>Examples</i>
Difficulties	Local producers experiencing a hard time	Uppsala still buying most products	Some consumers think that producers are enemies		FI3 <i>“I think most local food producers are in trouble, small but still trouble, it is difficult to compete with big brands or foreign trade, there are also many regulations, paperwork, finances that need administration, this makes it hard also to focus on cultivating the land.”</i> FI4 <i>“If I may tell you something, that I have always have in my head, is that some not all, but some people see us as their enemies, like we just want to take their money. I think, it would be nice to be like allies, at least small, medium farmers and consumers, I feel there may be something to see, a change of some sort.”</i>
Food Security	Sweden is food secure	food insecurity for some	Dietary choices	Media not helping	FI4 <i>“It is strange for me to think that Sweden or the County is not food Secure, but now, I am beginning to think that this insecurity may be occurring to some people. People with less, because crisis do not affect all people in the same manner, so I guess even if overall most people in the country or county are still eating well, some minorities may not be as fortunate.”</i> FI2 <i>“It is an interesting issue, this food security or insecurity, for me there is no question that Uppsala county, the region, Sweden are well in terms of food supply, but it is also true that people are experiencing trouble, economic trouble, we all are. But, I think we sometimes look at the wrong places. For example, It should be a good exercise to look at dietary choices. Why is people buying</i>

<i>Aim</i>	<i>Themes</i>	<i>Sub-theme 1</i>	<i>Sub-theme2</i>	<i>Sub-theme 3</i>	<i>Examples</i>
					<i>some exotic fruits coming from Spain or North Africa, and at a higher price, instead of buying local potatoes, which are also very healthy? I also think the media is encouraging in a bad way these trends in food choices, and is not good."</i>
Vulnerabilities	Risk in long supply chains	Terrible policies nor helping local farmers grow	European Market	Representation	F13 <i>"From my experience, too much regulations, there is a world of paperwork, accountancy, reports etc., that we need to keep in check, it almost seems that they do not want us to succeed, it seems the policies they elaborate to help us, are really to trample us. I can show you this letter that I sent months ago to LRF. It is just a small administrative matter, but no answer in 4 months."</i> F11 <i>"To depend so much on the European Market is not a good strategy, I think that we are importing more that exporting in the past few years. It seems a bit contradictory or? Heavy regulations, all these paperwork, reports it makes it difficult to keep up with foreign products, and at the same time these products make supply vulnerable in many points."</i>

Table 7. Devise solutions with producers

<i>Aim</i>	<i>Themes</i>	<i>Sub-theme 1</i>	<i>Sub-theme2</i>	<i>Sub-theme 3</i>	<i>Examples (FI =Farmer Interview)</i>
Ways to improve	Dietary choices	Citizen manifestation	Less regulations	More local farmers	FI2 “There should be a way to educate people about dietary choices, this definitely could help to spend less, buy from us local growers and at the same time eat healthy” FI5 “To start, ease up the regulations, change some of the policies, encourage the creating of new companies. Surely with less regulation or with policies that help entrepreneurs to create their own food business prices can go down and the local economy grow.”
Civil involvement	Efficient authorities	Structured Organization	Policy		FI3 “I do not normally speak about this, but I get the impression that we small farmers are not represented in an efficient manner. There are many laws or policies that are doing damage more than help”. FI4 “It all comes to how efficient and well-structured the authorities are, if they are not well organized, the policies that result from this bad process won’t be good for anyone, some of these policies seem to exist just for the sake of regulating”.
Authorities	Not to blame a specific Government but authorities in general	LRF	Lantmännen		FI1 “Is not use to blame past or present Government as a whole, but there are some functionaries that should be replaced by more capable people. People who really want to help improve the situation. In our farm, we have asked Lantmännen to provide us with assessment but their response time is around six months. That is

<i>Aim</i>	<i>Themes</i>	<i>Sub-theme 1</i>	<i>Sub-theme2</i>	<i>Sub-theme 3</i>	<i>Examples (FI =Farmer Interview)</i>
					a time that we do not have. So we had to spend money hiring a private advisor.” FI2 “I am part of LRF but I think they are more willing to help big farmers. There are many request that I have made to them, but still no answer. I can show you several letters during the past 4 months, but is like talking to the wind”.
Local food	Important but dwarfed by price	Benefits of localizing food production needs promoting	Local food production = Enhanced Local economy	Local food can bring together producers and consumers	FI3 “If people begging to buy locally, I guess that can also educate people about food and about what’s happing locally, maybe some of them will stop us considering us like rivals that just want profit and not provide healthy food for the community. FI5 “I don’t know if there are many people studying that, but for me is quite straightforward that local food has a direct effect on local economy. Again, this needs promoting from the authorities. The benefits of local food must be acknowledged and incorporated into well designed strategies, so that consumers are aware of how they can make the situation better, and also be prepared if things get more bad”.

5.3. Mental model

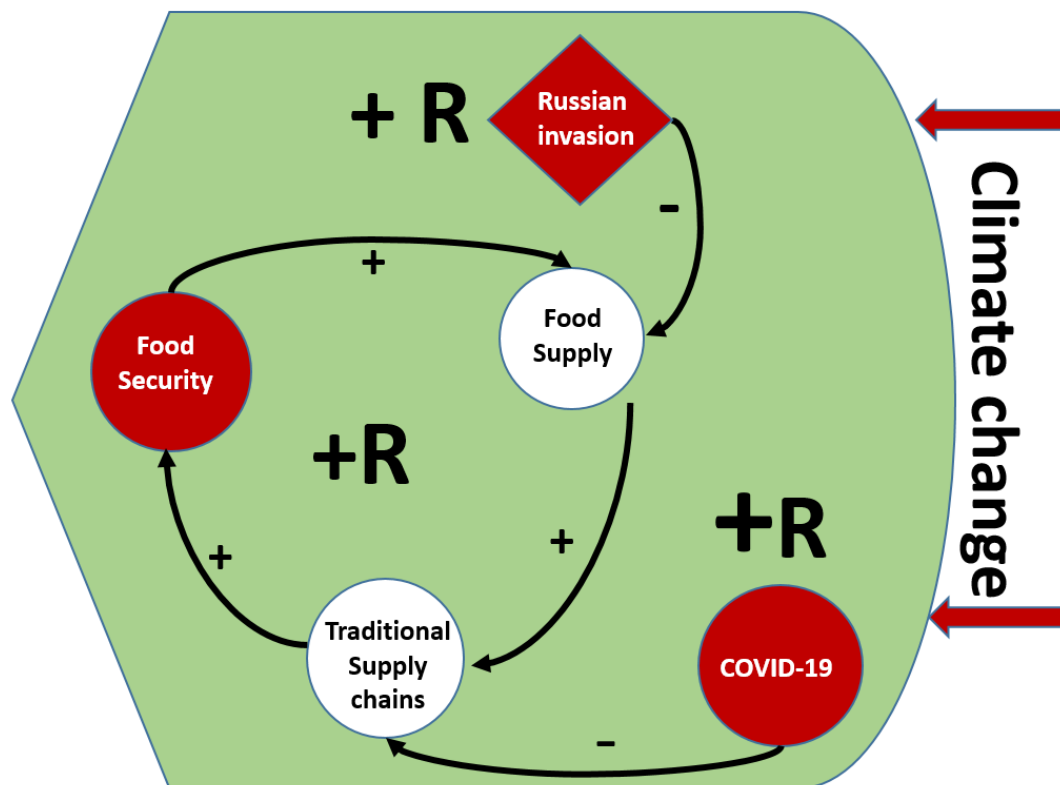


Figure 15. Primary Mental model establishing dynamics of the system

The mental models were produced during of stages three and four of the SSM, and it is important to consider that they are not a method that produced empirical data. The models are analytical tools that facilitated grasping the dynamics between the elements composing the situation being addressed. During the third stage, the situation deemed as problematic was conceived in terms of systems in order to be improved. In coherence with systems thinking, various boundaries were set, as a way to delimit the different interacting scales. It was also possible to consider these scales as different systems or sub-systems, or in other words, different levels within a hierarchy. For this work, this also meant to establish some of the dynamics and functions governing the systems.

Figure 15. Represents the preliminary understanding of the situation and its manifestations expressed in a graphic mental model. There, it is possible to note a connection composed of the elements: food security, food supply and traditional supply chains. The curved arrows signal the relations and the plus, and minus signs indicate the effect that one element has to the next one in the connection. Left alone, this relation results in a positive dynamic that favors food security. Food supply has a positive effect on traditional supply chains, which in turn has a positive influence on food security, which affects positively food supply, and the cycle starts again. The capital R with the plus sign within the loop constituted by food security, food supply and traditional supply chains means that this relation at the particular moment is a reinforcing dynamic that strengthens food security. However, this situation shifts with the incorporation of stressors such as, COVID-19, and the Russian invasion of Ukraine. Both this events plus climate change, which encompasses all relationships, have negative effects that transforms the possible outcome of the situation. After the interplay of these stressors, the connection between the elements of the model remain a positive, but in an escalating effect that is destructive for food security.

5.4. Refined mental model

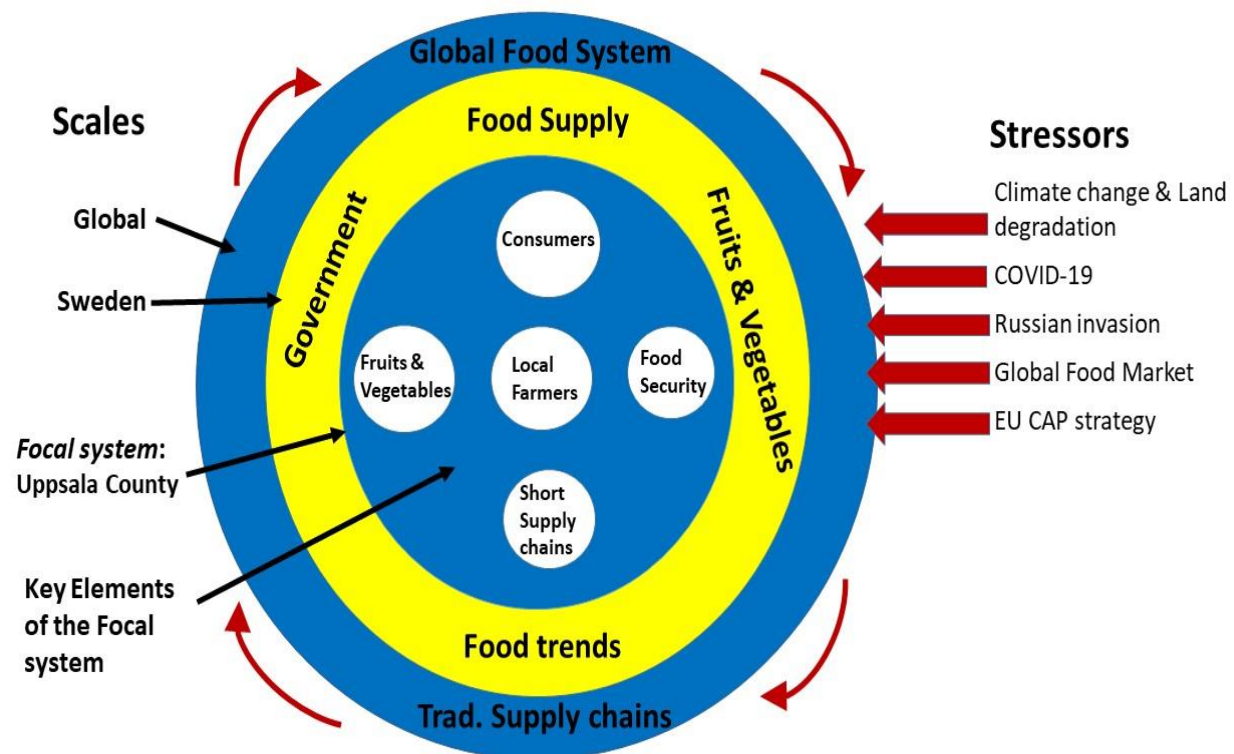


Figure 16. Mental model of the problematic situation

This version of the mental model relates to the 4th stage of SSM, where, as the process develops the model is improved. This model is also a result of what has been found in the previous stages and serve as an analytical tool to facilitate comprehension of the problematic situation. In figure 8, it is possible to grasp a more complex expression of the dynamics of the system. The circular representation and the curved arrows around the circumference try to provide the idea that system or situation is cycling. This comes from the notion that a situation is not static, and that the system was examined at a particular moment that will change with the designed actions to improve it. Moreover the circular nature also denotes the idea that change begins at the center, or at the smallest levels of society and then it quickly spreads to larger systems. Summarizing the model, first, there are three main scales (three circumferences) that interact with each other (Global, Sweden, and Uppsala County). There are also a set of elements particular to each scale, and that interplay between them. For instance, Government that is situated at Sweden level, also interacts with larger and smaller scales affecting them. In addition, outside the boundaries of the system (if there is such thing as outside) the stressors cause shocks that reverberate throughout the scales. A significant distinction at this point was determining Uppsala County as the focal system of study, as well, as identifying its most important internal aspects that relate to food security.

Table 8. Summary of the Problem-ranking

From the activity, it is clear than from their perspective the problem they are experiencing as graver is the situation related to food and beverages. This aspect, received the highest score in each of the six individual matrixes.

Perceptions on the situation	Food and Beverages	Health & Hygiene articles	Housing payments	Clothing	Leisure	Travel	Transportation	Rank
Food and Beverages	X	Food and Beverages	Food and Beverages	Food and Beverages	Food and Beverages	Food and Beverages	Food and Beverages	1
Health & Hygiene articles	X	X	Health & Hygiene articles	Health & Hygiene articles	Health & Hygiene articles	Health & Hygiene articles	Health & Hygiene articles	2
Housing payments	X	X	X	Housing payments	Housing payments	Housing payments	Housing payments	3
Clothing	X	X	X	X	Clothing	Clothing	Clothing	4
Leisure	X	X	X	X	X	Leisure	Leisure	5
Travel	X	X	X	X	X	X	Travel	6
Transportation	X	X	X	X	X	X	X	7

The problem-ranking matrix activity relates to the fifth stage of SSM of comparing the conceptual model with reality. This means to contrast stage four, which was the mental model constructed with stage two the problem expressed. The method was performed with a group of concerned participants from Dalaresan Street in the neighborhood of Sävja Uppsala, and at the same time, the exercise aimed to generate a discussion that developed afterwards during the focus group. During the process of the Problem-ranking matrix, I could revise if the previous understanding of the situation constructed by the individual interviews and both literature and statistical review reflected the reality of the problem itself. This comparison was the point at which intuitive perceptions of the problem were brought together with the mental model constructs.

In a narrative way the sequence of the activity developed, begun with me as a facilitator and the six persons who volunteered. At first, I presented the purpose of the activity and the overall topic. After this, the participants wrote what they considered problematic topics and sub-topics, from which they chose six. Then, I gave everyone a sheet of paper, and drew myself an outline of a matrix for reference. Then, they proceeded to elaborate each their own matrix incorporating the themes they had previously selected. After a while, they were ready with their rankings and then I quickly processed the values and did my own matrix summarizing the results from the six participants. It is important to tell that the result of the activity was discussed in the second moment of the participatory activities as the starting point of the focus group.

5.5. Focus group

The focus group began right after the conclusion of the problem-matrix ranking. The initial point of the exercise was a brief discussion that was recorded on audio concerning the results of previous activity. The reason for this was to stimulate healthy discussion and motivate the ideas of the participants. After this, I again acted as facilitator. I proposed each question aloud, and then each had a turn for answering, then there was time for discussion. This was done for every round of questions. It is paramount to mention that contrary to the plan, during the activity the participants proposed a novel and collaborative way to analyze the results of the focus group. In this manner, we came up with a “participative thematic analysis”. We briefly reviewed the most common themes that composed the content of the questions and discussions, afterwards the participants deliberated on which themes to eliminate. Then, after removing the themes that for them were not as crucial in content, they discussed each time the ones that were esteem as key. At the same time, they tried to construct in unison, meaningful phrases with the remaining themes. This I called “making sense” of the themes. The final product of the themes is the small message that is at the center of figure 6, which I produced for presenting the process in a condensed way the main ideas co-produced by the participants.

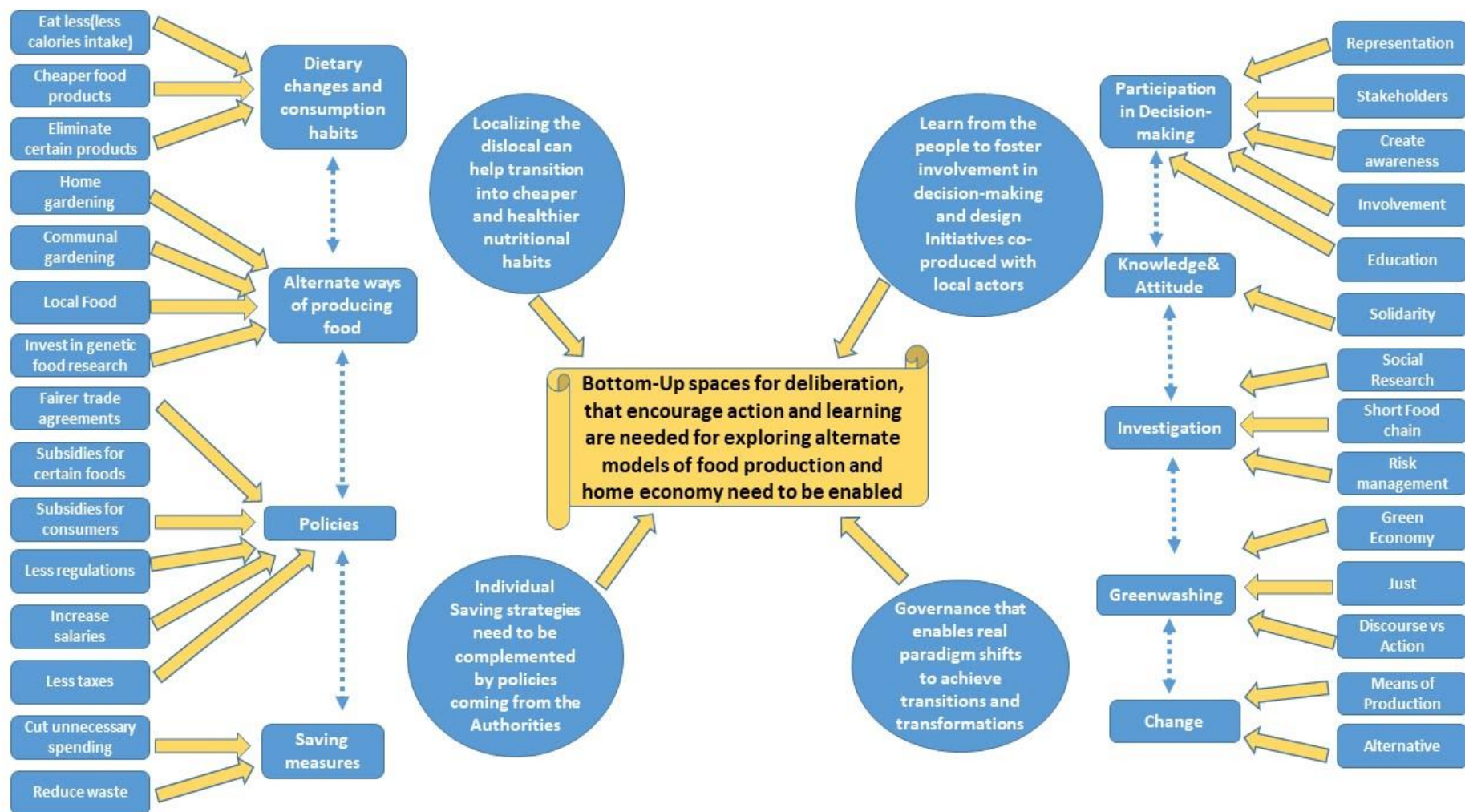


Figure 17. Summary of the focus group, presenting the main themes and messages derived from the activity

6. Discussion

The main purpose of my work was to provide a comprehensive discussion on food security through the lens of Uppland Region, integrating notions of local food systems, while enriching the process with the participative quality. From this stance, the aim of my Thesis was to address the problematic situation concerning food security in the region caused by worldwide land degradation the post COVID-19 era, and the Russian Invasion of Ukraine.

In this chapter, the results are first analyzed in terms of their individual meaning, and then, these are explained and synthesized as a whole in connection to the context, and to their significance in relation of the problem that this thesis is addressing. In the particular case of the mental models, due to their analytical nature in relation to the other methods employed, and to the integral comprehension of the problem, they are not scrutinized individually. It is also paramount to state that, even if my focal point is on Uppsala County, It is key to understand it, in the larger system, in the broader context. This is the reason why I kept contrasting information about Sweden. Also because there is much more data concerning the country as a whole. In addition, comparing these contexts, it is possible to grasp that many patterns, behaviors, dynamics that occur at country scale, reverberate to the Uppsala level, which means, that in a way it is impossible to detach one from another.

6.1. Analyzing the parts

6.1.1. Official documents and academic literature

The results are in a way obvious to interpret and they pushed forward my perception, that the situation perceived really exists, it is problematic and needs further research. The first relevant document that backed up my idea of the triple threat of worldwide land degradation, COVID-19, and the Russian invasion was the EU's initial response to Sweden's CAP strategic plan. The report acknowledges the integral link that these events have for food security. In this context, the EU also pointed out some relevant weaknesses in relation to Sweden's agricultural plans. This line of thought is consequent with investigations such as the one performed by Antonissen (2020) who highlights the risk concerning Sweden's supply chain, and how the country's high dependency on food imports augments food supply vulnerability. At the same Larsson and Sjöqvist, (2022), as well, as Ben Hassen and El Bilali (2022) alert that the current context in the near future will affect Countries like Sweden that are net importers of food.

Even if some literature supported my initial notion that the phenomena, which I have decided to refer as "the triple threat" is putting pressure on the global food system, thus creating risks for food security Sweden, I also discovered during the research process, that there is a gap regarding investigations that addresses these issues in relation to Sweden. I confirmed this notion with the work of Rost and Lundälv, (2021). Their work, which focused on reviewing academic literature concerning food security in Sweden, concluded that there is practically none. The necessity to fill the absence of research, is substantiated by a good quantity of reports by FAO (2022; 2023), IFPRI (2021) or World Bank (2023) that clearly exposed that food insecurity is worsening in Sweden, mostly affecting marginalized groups in disproportionally. This means that without proper investigations these issues have the tendency to escalate in ways that we do not yet understand or foresee.

6.1.2. Making sense of numbers

Examining the data gathered from the Swedish statistics database, as well as the graphics I produced, it was possible to comprehend using basic associational analysis, that concerning agriculture there is a shrinking pattern in Sweden and in Uppsala County. An important element with this method was that, since there are not many scientific articles studying Uppsala County or even Uppland region, a comparison between scales, (that is Sweden as a country and Uppsala County) was essential to implement for several variables. This corresponds to the increase in land prices per hectare both in Uppsala County, and in Sweden. This pattern is certainly increasing, as there is an escalating dynamic

acting. Higher land prices reduce the potential for further agricultural expansion. At the same time, the increase of global and regional European commerce seems to encourage this. Another phenomenon that strengthens this dynamic is the rise in the cost of agricultural production costs, the most evident examples are that of fertilizers and fuel. A result of this pattern seems to be the stagnation that micro and small agricultural companies have experienced in Uppsala County since 2010.

When reviewing organic agriculture, which is an important aspect of local food systems as a way to improve food security, I determined, that it has begun to decrease. This is noted in the decline of organic agricultural companies and organic agricultural land. Even if these contractions are currently quite small, they may escalate in the future, one argument for having this idea is what Rost and Lundälv (2022) mention, that the Swedish Government is more focused in food insecurity strategies beyond its borders. This means that most initiatives attempt to address food security in countries of the so-called global south, where issues of poverty, famine and malnutrition are causing severe damage to society. As far as the year 2019, Sweden collaborated with FAO through its international development cooperation agency with SEK 1.3 billion, and in the year 2021 alone a total of SEK 94 million. While these actions are not to be criticized, there is no clear budgetary increases for alleviating the situation, or enhancing preparedness domestically. Another reason for discerning that this trend may continue is the Swedish Pathways for Sustainable Food Systems (2022) which targets market mechanisms that further rely on international trade instead of securing a better degree of self-sufficiency. The consideration that food supply can be granted relying mostly on imports is leading to overlook other domestic issues that relate to food production. In my view, it is possible to associate this to the statistical findings that point out a relation between the increased imports, and an increment in the number of partner countries, which in turn seems to be inversely proportional to the spiking agricultural land prices, the contraction in total agricultural land, and stagnant agricultural companies. Another element that may increase vulnerabilities in food supply, mostly fruits and vegetables (where there is for some products 100% imports dependency) is the consumption of these by person, which according to data by jordbruksverket (2022) has increased 4% in the past three years.

6.1.3. What people say

The interviews with local consumers reveal that in their perceptions, it is difficult to label Sweden as food insecure, or to think that food security is at risk. At the same time, although it may seem contradictory, people also feel that, there is something wrong, that has changed or is changing. This perception is felt mostly in relation to what they eat and drink. They manifested, that for some reason not clear to them, prices have spiked, and that maybe, just maybe that has to do with war and the aftermath of the pandemic. On an interesting way, the people interviewed felt that the media is a bad influence, affecting in negative ways people's opinions on these topics. Most thought that the media weather exacerbates certain situations in disproportionate manners or softens other producing a certain apathy between the people. An interesting example that was brought forward is the notion that exotic diets are promoted by the media and in this way influencing to some degree peoples choices.

In regards to what consumers consider may be plausible solutions, there is not much room for interpretation, their answers manifest an urgent need for the authorities to design new measures that can put a stop the overall rise in prices. People agreed that for devising this plans, the Government should enable civil involvement, and create spaces for discussing issues related to food security. In this sense, not all were sure that localizing food production is the answer, but all consider that this idea should at least be studied and promoted more widely. In this particular point, my main takeaway is that consumers consider important to learn or to be educated about such alternate ways of producing food, because this could end up, in discarding a good approach out of ignorance. In addition, for some, another path to explore would be to just buy less, eat less, even if this means having fewer calories intake or a less nutritional diet.

6.1.4. What organic farmers say?

Organic farmers seem not to consider worldwide land degradation or climate change as an issue, and some even doubt that is happening in the way the media and politicians frame it. This perception originates from the fact that the negative effects of climate change are not felt in the local agricultural dimension as it happens in other regions of the world. In this sense, the producers that do consider climate change feel that is not affecting them, and that in the future it may be beneficial for them, because temperatures will increase and they will be able to cultivate a range of products that otherwise would be impossible to grow. When it comes to the producers that have doubts on how the media or politicians frame climate change, they feel that, since it does not hinder their business, they do not believe in it in practical terms. My reflection is that, if they were to feel the effects then they would believe it. These farmers also had the idea that climate change or land degradation as they prefer to call it, happens in other parts of the world due to poor management of resources, corruption, ignorance or internal conflicts. At this point is also essential to state that this is an individual finding, which does not necessarily mean that worldwide land degradation is not a vulnerability for Sweden and Uppsala County. What the farmers do believe is that COVID-19 and the War in Ukraine is affecting their business. They have felt the surge in production costs that may end up closing some of the companies. These higher production costs mean fewer incomes, and less capacity to grow or invest. To worsen their profit, people are buying less of their products. Simultaneously negative for them, an overwhelming set of regulations and paperwork hinders the efficiency to perform their work. In this matter, they state that organizations such as the National federation of farmers, Lantmännen or past and current Governments could do much more to facilitate solutions that integrate local small organic farmers. In this light, they feel that Governments in the last decade are favoring the European market, and discouraging local economy, which increases vulnerabilities in the food supply, due to long supply chains, that lack the nutritional and cultural value of local foods. Another interesting takeaway is that local producers also have the perception that the media in general is pushing consumers towards exotic diets, when they could eat cheaper and healthy food produced locally.

In concordance, among the actions that the interviewed farmers manifested as necessary, exist several that relate to Authorities. More specifically to reduce regulations, as a way that can stimulate entrepreneurs to invest in local small food business. In other words, a full revision of policies concerning local farmers should be done in favor of these type of companies. Coming back to the role of media, producers agree that it should promote the benefits of local foods, because they feel that there is a disconnection between local food and the people of the locality. They view local food systems as an educational path to bring together producers and consumers while enhancing the local economy.

6.1.5. Participative problematization

The main consideration that came from this activity was the evidence on how important the issue of the rising costs of food is for consumers. As presented in the Results chapter, all the participants ranked first food and beverages as the most problematic issue their everyday life. This was important in relation to other methods that point out that in people's perception Sweden is food secure, and that food insecurity is not close to happen. Also at this point, an apparent contradiction seemed to rise. People consider the acquisition of food as the major complication for their home economy, but they do not contemplate risks or major threats in the future of food supply, or food security for Sweden. The results from this activity also helped to contrast the mental models with reality, and strengthened the comprehension of the problematic situation.

6.1.6. Focusing on change

Besides the themes presented and the short messages produced during this activity, it is possible to give other meanings to the experience of the focus group. The additional significance that I view, is that, even if there is an apparent increased in apathy, and lack of civil involvement regarding issues of food security, people are willing to take action, to collaborate, and are open to new ideas, that are

different from their established beliefs. For instance, most participants did not know of the benefits of localizing food production, but they were eager to learn about it, and ponder it as a possible solution to correct the problematic situation encompassing food prices. These signs show that people are waiting for similar activities to manifest their ideas, not only in passive way as a mean to express themselves, but to engage in transformative actions. This indicates that there is a certain need for initiatives that enable spaces for civil organization in bottom-up processes of decision-making. In addition, even if my research would not end in a policy change, they were concerned enough to propose an ingenious participative manner to analyze, and process themselves the themes from the discussions.

6.2. Synthesizing

The synthesis of the results helps to discern interconnectedness between them. At the same time, looked through the lens of the theories and frameworks describe in that chapter, these give meaning to the findings as a whole in the context they occurred, as well as, their significance for the problematic situation examined. Additionally, the integrated interpretation of the results is here presented as a mean for fostering the discussion, and provide inputs for answering the Research questions.

6.2.1. Food security is not risk secure

For entering the discussion and its elements, it is important to approach it in connection to the research questions.

1) What vulnerabilities exist is the current situation of food security in Uppsala County?

Making sense of the integrated results, I have found several relationships between the individual findings that translate into vulnerabilities. First, it is critical to consider that even if my thesis was not designed in terms of analyzing discourse; I discovered a dominant discourse both in the academic and civil communities that favors the idea that Sweden is food secure. I do not pretend here to refocus my thesis to discourse analysis, but the repetition and persistence of the same message is not possible to omit, and it has made me reflect on this constant equation of food security = Sweden. From this perspective, I will not engage in attempting to unveil what the discourse means in its totality or trace its origins, but I definitely feel an imperative duty to disclose it as an important discovery, which could be a potential source for vulnerabilities. In the case of consumers, this can be noted from the interviews and focus group, where most participants express discomfort, and disbelief at the idea of Sweden being food insecure or at the notion of the country approaching such at state in the near future. Regarding academics, the preliminary literature review supported the belief that countries in the global north, such as is the case of Sweden are food secure. Studies as the one performed by Borch and Kjærnes (2016) or Rost and Lundälv (2022) point out that since 1975, little knowledge has been produced that addresses Food Security or Insecurity in Sweden. This gap in academic research means in simple terms, that the academic discourse has remained the same. Certainly, from my action research stance, this dearth of learning and knowledge production represents the main root for vulnerabilities, and an obstacle for taking action.

In other words, I argue that a lack of constant learning, knowledge production, and a dominant perception of Sweden being food secure is the origin of risks and vulnerabilities for the country and Uppsala county in several ways, that tend to overlook underlying issues and challenges that need to be addressed. My findings support this, first due to the lack of knowledge that the consumers and producers had in relation to take measures for improving their home economy. Second, there seemed to be no learning in terms of adequate ways to mitigate the constant rise in food prices, in terms of nutritional value, or in terms of how their dietary choices affect their cost of life and health. In addition, considering the almost nonexistent creation of knowledge observed in the exploratory literature review researching food insecurity in Sweden, it is possible to recognize that the perception, that Sweden or Uppsala county are food secure hinders its capacity to conceive actions to improve the situation or to enhance preparedness. I noted this in the gap of knowledge that most consumers have regarding the potential benefits of localizing food production. This was exposed during the interviews

and focus group, where at a first the participants did not think that that local food from the county could strengthen food security. However, after engaging in conversations with them, they exercised self-reflection, where they felt that in reality they do not know enough about this alternate way of food production. During the focus group, the two participants that had some knowledge of local food systems, quickly and spontaneously began teaching to the others what they knew. It was a simple process of active learning, that revealed that learning and the co-production of knowledge is a lack and a must. Furthermore, exploring some of the vulnerabilities inherent to the dominant perspective of Sweden's food security, is possible to highlight that, there is a neglecting of vulnerable groups. A lack of discussion and mutual learning gives the perception of overall food security in Sweden, and it is leading to the oversight of marginalized groups within the population. It creates a false sense of security, overshadowing the fact that low-income individuals, marginalized communities, or immigrants, may still face food insecurity. This was mostly exposed during the interviews to consumers, where the different backgrounds of the interviewees manifested into contrasting views on how food prices were affecting their home economy. Those that were long-term migrants or had low income jobs (or part-time) felt stressed and uncertain towards their future. In two cases, the interviewee manifested that the family had already been eating less quantities and not so nutritional foods. On the contrary, people that were from Swedish origin and had well paid jobs did not feel that the spike in food prices meant a threat to their home economy. These findings are definitely in concordance with the World Bank (2022) data that showed a persistent increase in mild-to-severe hunger in Sweden since 2015. Furthermore, FAO polls (2022) produced for their yearly report uncovered this same behavior, adding the fact that marginalized groups composed mostly of immigrants are the ones feeling these increments in food insecurity. Additionally, to support the findings that this trend will continue, some of the latest FAO reports (2023) that address implications of the Russian war on Ukraine predict that the effects of this situation will worsen, deteriorating the global food system.

Coming back to potential vulnerabilities originated by lack of constant learning, knowledge production, and a dominant notion of Sweden being food secure, I also argue that it can lead to ignore structural inequities within Uppsala's food system production. Issues such as unequal access to resources, disparities in income and capacity to assimilate regulations can turn into barriers for producing nutritious, cheap food. Ignoring these structural inequities can augment existing inequalities and hinder progress toward a more inclusive and equitable food system. To support this notion, I go back to the local small organic producers, that manifested in the interviews that for them it is hard to compete with international brands, and that overwhelming regulations, as well as, land prices make it difficult to maintain organic labels. In their perspective, these regulations are easy to carry for big farmers, but it becomes disproportionate in the case of small or micro local producers. To complement this notion, the quantitative statistical data gathered, reveals an interlinkage between (1) incrementing agricultural land prices, (2) increase in agricultural inputs, (3) a persistent decrease in the total agricultural land, and (4) a stagnant dynamic when it comes to the creation of new small or middle agricultural companies. In simple terms, with the current regulations and land prices, it is very difficult for new local small farmers to appear. This is coherent with what Nilsson (2009) concludes in her investigation, that the current agricultural structure has potential risks in a twofold manner. First by over relying on long supply chains that can be disrupted and second by creating inequalities in the food production system, that affect mostly local small farmers.

In addition, an insufficient learning and knowledge production regarding Food Security in Sweden that challenges the official perception that Sweden being food secure can lead to complacency from authorities and the citizens. This in turn can cause inadequate preparedness for shocks and disruptions in the Food System. I could note this mostly during the interviews with producers. They expressed that for small businesses, the current policies do not seem to match the reality. In their view, current regulations seem designed to discourage small producers, instead of providing them with the means to develop, and strengthen their companies. At the same time, I learned from the consumers (in interviews, problem-ranking matrix, and focus group) that they were not aware of the impending vulnerabilities or risks within Uppsala's food supply. Here I agree with Sonnino (2018) who argues that this lack of awareness from consumers, translates into a lack of discussion, thus turning also into

complacency from both Government and citizens. This also causes a limited focus on alternative food production practices. To support this idea, I take a note from the EU initial response to Sweden's CAP strategic plan (2022) that highlights the perception that Sweden's food security measures may disregard the need to prioritize alternative agricultural practices, resource conservation, and environmental stewardship. Sweden's reliance on imports for a significant portion of its food means that the environmental impact associated with production, transportation, and packaging is often overlooked. (Horn, Ferreira and Kalantari, 2022).

This connects to an issue that has been highlighted in every result from the data gathering process, which is dependence on global markets. The statistical data gathered shows that for Uppsala County, imports help to meet the demand for a diverse range of food, but a more in-depth scrutiny of the situation as (Sonnino, 2013)) suggests that this overreliance on global markets exposes the country to potential disruptions, such as trade restrictions, price fluctuations, or supply chain vulnerabilities. Choosing a stance in favor of increasing local food systems, my interpretation, of the statistical data, previous studies, and stakeholder's perceptions, concurs to the idea that focusing on global market mechanisms to solve domestic food supply is in fact a vulnerability. Certainly, my stance does not oppose the benefits of regional or global food systems, from my perspective it becomes a vulnerability when implemented disproportionately in comparison to local production.

6.2.2. Localizing the dislocal

This section of the discussion aims to the second Research question.

2) To what extent can food security improve by localizing production in Uppsala County?

A study by Brodt et al., in 2013 revealed the inherent benefits that localizing production has for Food Supply. Their study investigated the specific of tomatoes; it contrasted supply in terms of regional and local production. Among other aspects, it concluded that local production of tomatoes improved the stability, reliability, traceability and quality of the tomato supply. It supported food security, reduced environmental impact, fostered local economic growth, and provided consumers with fresh and flavorful tomatoes throughout the year. In the case of Uppsala County I argue based on statistical data that there is potential for many products, as could be the case of potatoes, apples, oats, carrots, cherries, plum, or pears. These are just a few examples of products that have been grown in the County during the previous decade, and that its production could be incremented. I consider that in turn, this could decrease reliance on imports, making Uppsala County less vulnerable to disruptions in the regional or global food supply chain. Additionally, by producing a larger portion of its food domestically, the County can enhance its food security and ensure a more stable and reliable food supply for its population.

Another aspect that came to my understanding when revising the mental models produced as analytical tools is that; Localizing food production can reduce shocks that stressors such as what I have called the triple threat may inflict on the county's food supply. Additionally, it would reduce its dependence on external drivers such as international trade, global market fluctuations, and geopolitical tensions. This reduces the risk of supply chain disruptions, trade restrictions, or price volatility, thereby enhancing the country's control over its food supply. This line of thought is also supported by the considerations made by the EU in regards of Sweden's CAP strategic plan. The commission motivates Sweden to balance its dependency on food imports, by incrementing local production and suggests increasing incentives to local small organic farmers. This as a way to address the current and future implications that climate change and the Russian invasion of Ukraine have for Food security.

An interesting element that came from the interviews and focus group is fresher and higher-quality food. From the participants perspective local production allows for the delivery of fresher, thus more nutritious food. Returning to the study by Brodt et al., (2013), it explains this idea, by arguing, that since locally grown food products have shorter transportation distances, they can be harvested at peak ripeness and delivered quickly to markets and consumers. According to their study, this maintains the

nutritional content, and overall food quality, which usually does not happen with long supply chains. Against this claims, there are also some authors such as, Galli & Brunori (2015), who claim that the global market with its traditional or long supply chains is so well developed, that products coming from far regions are put into local food stores faster, cheaper, and with a high degree of traceability. To contrast this I argue in favor of research conducted by Edwards-Jones et al. (2018) that disclosed that Localized production promotes greater transparency and traceability in the food supply chain. Consumers have better access to information about the origin, production methods, and handling of their food. This enhances food security, reduces the risk of contamination or adulteration, and enables consumers to make informed choices about the food they consume.

One of the most debatable aspects for this discussion, which originates from my initial revision of academic literature and official documents relates to the notion that local food systems or localizing production has a positive effect on the local economy. Even if this controversy did not arise from the empirical material, I find it important to relate the discussion to issues raised during the first phases of the thesis, and expand or refine what was previously expressed. Regarding the detractors of the statement that local food is inherently beneficial for the local economy, it is possible to mention authors such as, Galli & Brunori (2015), Low et al. (2015), Deppermann et al. (2018) or Kneafsey et al. (2013). These researchers argue that there is no evidence of the economic benefits of locally grown products. They argue that the Economic value is highly constrained by two factors. First that the demand for local products may be limited by the number of consumers who can afford local products, or who are willing to do so, which can eventually decrement sales due to low consumer interest. Second Due to the small scales nature of local systems and the managing of inputs through shorter supply chains, local producer's profitability may also be constrained depending on how much they can reduce their production costs. In my Thesis the consideration of local food as beneficial to local economies derives from the literature reviewed some examples are studies by the University of Leeds (2018), (O'Neill, 2014), Nilsson (2009) or Feenstra (1997). In these studies, local food production supports the local economy by creating jobs and fostering economic growth. It benefits farmers, food processors, distributors, retailers, and other related sectors, stimulating economic activity within the region. Moreover, a study by Nilsson (2006) demonstrated that Farmers involved in a local food initiative manifested, that a larger percentage of the profit ended up with the farmer/producer instead of retailers. This also lead to more money being generated for the community and staying in the community as members of local systems use their money within the local area, for example, trading with local suppliers.

6.2.3. A note on Learning

A final element, which is crucial and elusive to quantify, is the positive impact that localizing food production has for learning, for educating both consumers and producers. According to the local producers interviewed, local food production often involves opportunities for citizens to visit farms and participate in open houses or farm tours. I argue that these experiences provide firsthand exposure to agricultural practices, allowing individuals to learn about crop cultivation, livestock rearing and sustainable farming methods. Farm visits offer valuable educational experiences for people of all ages, and backgrounds fostering a deeper understanding of where food comes from, and how it is produced. Another example of how local food systems foster learning and the co-creating of knowledge are community gardens. During the focus group, two participants expressed that urban agriculture projects may provide opportunities for citizens to actively engage in food production. According to Barron (2017), these initiatives can be educational platforms where individuals can learn about gardening techniques, plant cultivation, and sustainable practices. Participants gain practical knowledge, develop gardening skills, and deepen their understanding of the agricultural process. At the same time, workshops and training programs can be designed to enhance the learning process. From the producers perspective it is possible to create networks and organize workshops, and skill-sharing sessions for citizens interested in learning about farming, gardening, food preservation, and other related topics. These educational initiatives provide a platform for individuals to acquire new skills, share knowledge, and engage in hands-on learning experiences.

In addition, as discussed during the focus group, localizing food production can have a significant impact on education within schools. School gardens enable students to participate in planting, nurturing, and harvesting crops. They offer potential interdisciplinary learning opportunities, integrating subjects like science, biology, environmental studies, and nutrition into practical experiences. Additionally as Feenstra mentions (1997) the creation of farm-to-school programs connect students with local farmers, promoting awareness of local food systems and encouraging healthy eating habits. Another quality is that local food production often involves community events and farmers' markets, where citizens have the chance to interact directly with food producers. Seyfang (2006) manifests that these events provide a space for knowledge sharing, where farmers can educate consumers about sustainable farming practices, food sourcing, and the benefits of locally grown food.

Lastly, a brief synthesis of some benefits that organic producers expressed is that, local food production encourages a deeper understanding of environmental issues and the importance of conservation. Consequently, by emphasizing sustainable farming practices, such as organic farming, water conservation, and soil health management, citizens can learn about the impacts of agriculture on the environment and the importance of sustainable food production. These perceptions from the Uppsala county producers are coherent with Ernwein's (2014) view, that this type of learning and knowledge promotes responsible consumption and environmental stewardship. Through these educational opportunities, citizens gain a deeper understanding of the food system, fostering appreciation for local agriculture, sustainable practices, and the connections between food, health, and the environment. This has the potential to empower individuals to make informed choices, promotes sustainable lifestyles, as well as, strengthens the overall knowledge and awareness of the community.

7. Conclusions

In this chapter, the research questions are answered in a direct manner, and then a final reflection that relates to SSM and action-research is presented. An Additional aspect concerning learning, which is essential to assimilate is that practicing SSM from an action research stance calls for finding insights for conclusions throughout the whole research process. In practical terms, this means that conclusions and answers to the research questions may not come exclusively from the empirical material, but may originate in the preliminary structuring of the problematic situation.

7.1. Conclusions regarding the first research question

1) What vulnerabilities exist in the current situation of food security in Uppsala County?

1. Uppsala County has significant number of small-scale farmers who may face challenges such as limited access to resources, overwhelming regulations, financial constraints, market competition, and changing consumer preferences. These factors can affect their ability to sustain agricultural production, potentially damaging the availability of locally produced food.
2. Income inequality and affordability are deteriorating food security in some segments of the population in Uppsala County. These consumers may face economic challenges that hinder their capacity to access adequate nutritious diets. In connection to this, high food prices, low wages, unemployment, poverty, and social inequalities can also contribute to food insecurity within the County.
3. A lack of diversity in the food system can make it vulnerable to shocks and disruptions. In the producer's perspective the food system in Uppsala County heavily relies on a few key crops or imports, any issues such as disease outbreaks, pests, or trade disruptions affecting these crops can have significant impacts on food availability and access.
4. Infrastructure and supply chain vulnerabilities are regarded in the consumer's perspective as weaknesses that create further vulnerabilities in the food system, such as insufficient storage facilities, transportation infrastructure, and distribution networks. This in turn damages the reliability of the food supply chain, potentially leading to food waste, spoilage, or delays in food distribution.
5. Persistence of a well-established notion from consumers and producers that Sweden is food

- secure even if they feel that their home, and business economies are quickly deteriorating.
6. Lack of learning and knowledge from the producers regarding the effects that worldwide land degradation has for food security in Sweden.
 7. Lack of learning and knowledge from the consumers regarding measures that can help mitigate the effects that the constant spikes in food prices is having for their home economy. Learning and knowledge also play an important part when selecting certain expensive exotic dietary choices that are not always the most nutritious options.
 8. Dearth of spaces that foster civil involvement and that facilitate the co-creation of knowledge between citizens by mutual learning, sharing and discussions.

7.2. Conclusions concerning the second research question

2) To what extent can food security improve by localizing production in Uppsala County?

Localizing food production in Uppsala County has the potential to increased food self-sufficiency, and reduce its dependence on external food sources. In addition, by cultivating a diverse range of crops and rearing livestock locally, the County can enhance its capacity to meet a significant portion of its food needs. This strengthens preparedness along the global food supply chain and enhances the County's ability to provide food for its population. Furthermore, by having shorter supply chains transportation and store requirements decrease, ensuring that food reaches consumers more quickly and efficiently. Additionally, localizing food production combined with adequate policies can stimulate economic development and create employment opportunities within Uppsala County. It supports local farmers, distributors, retailers, and other related sectors, contributing to economic growth, as well as, enhancing the livelihoods of individuals. Finally, local production often emphasizes on alternate agricultural practices such as organic farming, regenerative agriculture, and reduced chemical inputs. However, it is important remember that complete self-sufficiency may not be feasible for all food items due to factors like climate limitations or resource availability. Therefore, a balanced approach that combines local production with responsible global trade can help achieve a robust food supply and security.

7.1 Final Reflection

This section deals with learning as a transversal element during the Research process, it is a reflection that I, together with my supervisor for this thesis agreed as important to include. We consider that, conscious reflective learning is the most important aspect of this process that is called master's thesis. From this stance, the product of the research process is learning, and not the document produced for the thesis course. In addition, from the action-research stance, learning can be reflected considering three aspects. (1) Learning about me (how the process has changed me). (2) Learning about the situation, and (3) Learning about the Methodology.

7.1.1. Learning about me

During this process, I have learned that I have a strong tendency to try to accomplish more than time allows, or more than is necessary. This can be easily noted in the many methods and techniques that I used in order to produced results. At the same time, I have felt a strong commitment to engage in a problem-oriented approach that could really foster change or lead to actions. I struggled with my inner self during this research process, because I needed a constant reminder that my work would not achieve the desired state of the situation, and that I would have to be happy with attempting to address the problem, which in SSM terms is to devise preliminary actions that could improve the situation. In addition, it is most important to say that, by engaging with consumers and producers in interviews that were more like conversations, I learn about my capacities in terms of communication skills. I have is no doubt that this skills have further developed. Furthermore, from being close to the people, I have reinforced my conviction of working with real world problems. I now that I want to pursue projects that involve problem-oriented, action-oriented or learning-oriented approaches. In addition, I learned to go back into the process and change, fix certain aspects that were not correct. For me looking back

at a process, and acknowledge that I make mistakes that needed addressing, is hard, but during this thesis I feel that I have finally embodied this process in a more natural way.

7.1.2. Learning about the situation

Even if the timeframe for achieving an in-depth learning about the situation was short, I could say that I learned and I will continue to learn. I feel the most essential aspect that I embodied was how eager people are to be involved in participative activities, to engage in healthy discussions concerning issues that affect their everyday life. Even if most people do not consider that Sweden may be food insecure, they were open to the idea of discussing the possibility, because they were aware of lack of knowledge in relation to these topics. This means that they are not just willing to discuss, but learn from the situation, and from each other. I felt a similar experience regarding the eagerness that consumers and producers had in relation to learn more about how localizing food production might contribute to strengthen preparedness. At that moment, they were unsure of its benefits, but at the same time, they acknowledge that they did not know enough, that they needed relearning, education on this alternative.

Among some of the actions that the participants from individual interviews and participative activities devised I can mention a few: Dietary changes and consumption habits, Alternate ways of producing food, revision of current policies, and design of new ones, more participation in decision-making, foster change in people's knowledge and attitude. This also translated into more learning about the situation, in this case the condensed lesson coming from the actors' perspective involved is, that Bottom-Up spaces for deliberation, that encourage action and learning are needed to be enabled for exploring alternate models of food production and home economy. In other words, the situation can change, first if decision-makers enable a combination of civil involvement and the creation of spaces for discussion. Second, if as an outcome of the previous process a set of actions devised by all stakeholders are set in motion. Third, a constant review of the situation takes place.

7.1.2. Learning about the methodology

This was probably the hardest part for me to reflect on. As a novice practitioner of SSM, I learned to value its flexibility in terms of methods, techniques, activities. I felt that given a longer timeframe SSM can really address problematic situations, and achieve transformative actions that with the necessary iterations can improve the situation. I learned that SSM with its seven stages might appear a prescriptive or normative approach, but is not, because for every context, for every situation, the problem, the problem, the actions, thus the process is unique. An important insight in relation to the Methodology was the justification of using a soft systems approach instead of hard systems. I consider of vital importance to ponder the quality that SSM has when solving real-world complex situations that involve human activities, as well as, its connection to action-research. Additionally, I learned that for future investigations, I would need to grasp better how to integrate SSM with a mixed-methods research design. For now, I feel that SSM may have been enough framework for piloting my work, but I wanted the strength of both. Certainly, this also signifies that my learning should never stop, and that it requires great amounts of practice.

8. Limitations

1. Limited sample size: Since my study is based on interviewing a limited number of consumers and producers, the findings may not be representative of the population. In addition, with a longer timeframe and budget a greater quantity of producers could have been interviewed. This could have potentially provided better insight into their diverse perspectives. Moreover, the aforementioned restricted the number of focus groups to one, thus constraining the participative action-research quality that Thesis intended.

2. As the investigation focus on the Uppland region, the results, indications and conclusions derived from the study are not necessarily valid to other parts of Sweden. This is consequent with the non-

positivist approach that this Thesis wants to follow, which leans to the constructivist paradigm where knowledge is context dependent (Freire, 1978: 58; Ackermann, 1982).

3. Since time is a limiting factor, my Thesis addresses the problem but it does not reach the transformative dimension that would mean reaching a solution. For “fixing the problem”, several “phases” or “stages” are required until the situation improves from being problematic to the desired state.

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Appendix A: Interview guide

Version for consumers

Introduction

- 1- Introduce the purpose of interview, the topic, and myself. (Displaying my credentials as part of a master's programme from SLU and UU).
- 2- Explained the utilization of the data, and its confidentiality
- 3- Informed consent (Having informed of the previous two points, ask for the explicit permission to conduct interview.
- 4- Ask if it is possible to record the interview (sound only)
- 5- Considering COVID-19, The Russian war and the effects of climate change or land degradation. Have you experienced difficulties in your business? If so, what kind?
- 6- Do you think other farmers or people in Uppland have experienced or are experiencing difficulties in relation to these phenomena?
- 7- Do you know other farmers or people that have experienced this?
- 8- Do you feel that the situation is improving?
- 9- How are you dealing with the increment in agricultural inputs and fuels?
- 10- If farmer rents land: How are you dealing with the increment in rent price for agriculture?

Food Security

- 11- How have you experienced the rise in food prices?
- 12- What do you understand as Food Security?
- 13- Do you think Sweden may experience Food Insecurity?
- 14- If not. In your perspective, what would it take for Sweden to become food insecure?
- 15- If so. How do you think this is happening?
- 16- Do you perceive any problem in Sweden or in Uppland in relation to Food?

Vulnerabilities

- 17- Do you think that the Swedish food production system has become more vulnerable over time? (Time relation varied depending on the person's age)
- 18- If not. Do you think it may be in the near future?
- 19- If yes. Do you have any opinion as to why or how these vulnerabilities came to be?
- 20- Do you think that relying on the global or European market for food supply increases vulnerabilities?

Ways to improve?

21- Do you feel that concerning Food Security, vulnerabilities or possible risks, that there are aspects which need improving?

22- If so. Elaborate.

23- Do you think that there is lack of civil involvement?

24- If so. What in your opinion causes this?

25- In your view, is there value in having more civil involvement in co-designing policy or regulations when it comes to food security?

26- From your perspective, what could be done to get people to participate or to actively engage?

27- Do you think authorities (local, or country level, LRF) could be doing more to improve the situation?

Local food production

28- Where do you sell most of your products?

29- Is it difficult to sell locally? (To Uppsala)

30- Or do you feel that it is easier to sell outside the region?

31- Do you think that incrementing local food production can improve the current food situation?

32- Do you feel that it was easier in past decades to establish a local food farming business?

33- If so. Why do you think, it has become difficult?

34- Do you have an idea on how new local small food producers could emerge?

Interview guide: Version for producers:

Introduction

1- Introduce the purpose of interview, the topic, and myself. (Displaying my credentials as part of a master's programme from SLU and UU).

2- Explained the utilization of the data, and its confidentiality

3- Informed consent (Having informed of the previous two points, ask for the explicit permission conduct interview.

4- Ask if it is possible to record the interview (sound only)

5- Have you felt a strong increment in the cost and/or quality of living?

6- If so. Can you mention and rank these aspects?

7- Considering COVID-19, The Russian war and the effects of climate change or land degradation. 8-

Have you experienced difficulties in at home? If so, what kind?

8- Do you think local farmers or people in Uppland have experienced or are experiencing difficulties in relation to these phenomena?

9- Do you know people that have experienced this?

10- Do you feel that the situation is improving?

11- In your opinion what could be done to improve the current situation? (As a citizen, and if you were part of the Government)

Food Security

12- How have you experienced the rise in food prices?

13- Do you perceive these increments as a threat for your home economy?

13- What do you understand as Food Security?

14- Do you think Sweden my experience Food Insecurity?

15- If not. In your perspective, what would it take for Sweden to become food insecure?

16- If so. How do you think this is happening?

17- Do you perceive any problem in Sweden or in Uppland in relation to Food?

Vulnerabilities

18- Do you think that the Swedish food production system has become more vulnerable over time? (Time relation varied depending on the person's age)

19- If not. Do you think it may be in the near future?

20- If yes. Do you have any opinion as to why or how these vulnerabilities came to be?

21- Do you think that relying on the global or European market for food supply increases vulnerabilities?

Ways to improve?

22- Do you feel that concerning Food Security, vulnerabilities or possible risks, that there are aspects which need improving?

23- If so. Elaborate.

24- Do you think that there is lack of civil involvement?

25- If so. What in your opinion causes this?

26- In your view, is there value in having more civil involvement in co-designing policy or regulations when it comes to food security?

27- From your perspective, what could be done to get people to participate or to actively engage?

28- Do you think authorities could be doing more to improve the situation?

Local food production

29- How much effort do you put into buying local grown food?

30- Do you consider the potential benefits of buying local food from Uppland region?

31- Which is the determinant factor in buying food, prices? Organic certification? Locally produced?

32- Do you feel that your dietary choices affect your perception on the current food situation?

Focus group main Questions:

Do you think that it is possible to take actions that will improve the current situation in terms of the food you acquire?

Have you considered actions that can help you with your personal or home economy situation in terms of the food you acquire? If yes what are those actions or measures?

How could the authorities improve the current situation?

What do you think of Uppsala's food system?

Do you think that food prices can be improved? If yes. How can this be done?

Have you thought of buying more local food?

How do you think authorities could help to increase Local Food production?

What do you think can be done to foster civil involvement concerning issues of Food Security and/or Food supply?

Appendix B: Complementary statistical data

Statistics for crops in Uppsala county, share percentage of organically produced crops, number of organic companies, and comparison of organically grown land in the year 2021 vs 2020.

Crops Uppsala (organic and conventional)	Share of total organic land area 2021	Organically produced crops	Organically grown land	Number of Organic companies 2021
Winter wheat	8.3% (decreased 7% vs 2020)	Winter wheat	Increased 4.3% vs 2020	112 (increased by 7 companies vs 2020)

Spring wheat	14% (increased 52% vs 2020)	Spring wheat	Increased 35% vs 2020	43 (increased by 3 companies vs 2020)
Rye	32.8% (decreased 2% vs 2020)	Rye	Decreased 44% vs 2020	21 (decreased by 13 companies vs 2020)
Autumn grain	15.2% (decreased 20% vs 2020)	Autumn grain	Decreased 18% vs 2020	5 (decreased by 1 company vs 2020)
Spring grain	6.9% (increased 13% vs 2020)	Spring grain	Decreased 5% vs 2020	66 (decreased by 15 companies vs 2020)
Oats	29.5% (increased 8% vs 2020)	Oats	Decreased 8% vs 2020	109 (decreased by 9 companies vs 2020)
Rye wheat	No recent data	Rye wheat	Decreased 48% vs 2020	13 (decreased by 2 companies vs 2020)
Spring rye wheat	31.8% in 2020 (no recent data)			
Autumn rye wheat	25.3% (decreased 46% vs 2020)			
Mixed grain	61.1% (increased 13% vs 2020)	Mixed grain	Decreased 19% vs 2020	17 (decreased by 2 companies vs 2020)

Maize	0% in recent years	Maize	1 hectare	4
Total grains	11.3% (no change vs 2020)	Total grains	Decreased 3% vs 2020	178 (decreased by 2 companies vs 2020)
Peas	5% (decreased 42% vs 2020)	Cook peas, fodder peas etc	Increased 7% vs 2020	58 (increased by 6 companies vs 2020)
Field beans	96.4% (increased 13% vs 2019)			
Fall trap	3.7% (decreased 30% vs 2020)	Fall trap	Decreased 9% vs 2020	11 (increased by 2 companies vs 2020)
Spring rape	No contribution	No contribution	No contribution	No contribution
Winter rape	58.2% (decreased 7% vs 2019)			
Spring turnip	0% in recent years			
Oljelin	0% in recent years	Oljelin	None in recent years	None in recent years
Straw seed crops for green fodder	95.6% (increased 64% vs 2020)			
Maize for green fodder	No contribution			

Potatoes	12.3% (increased 6% vs 2020)	Potatoes	Decreased 13% vs 2020	12 (decreased by 4 companies vs 2020)
Autumn currants	No data	Autumn currants	Increased 5% vs 2020	16 (increased by 1 company vs 2020)
Total canola and turnip		Total canola and turnip	Decreased 0.5% vs 2020	25 (increased by 2 companies vs 2020)
Other plant species	No data	Other plant species	Decreased 3% vs 2020	46 (decreased by 4 companies vs 2020)
		Total arable land	Increased 1.4% vs 2020	257 (increased by 1 company vs 2020)
		Pastures	Increased 2.4% vs 2020	185 (increased by 5 companies vs 2020)

Source: jordbruksverket, 2021

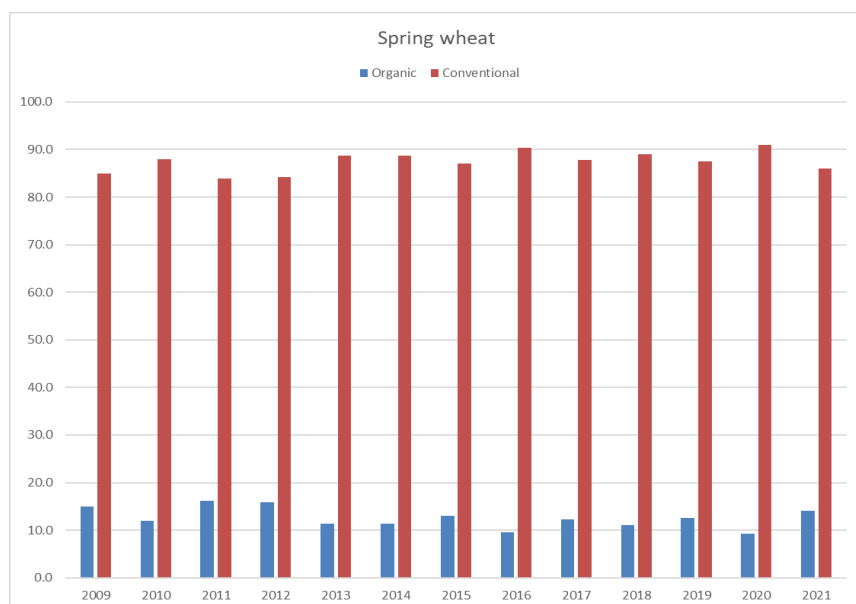
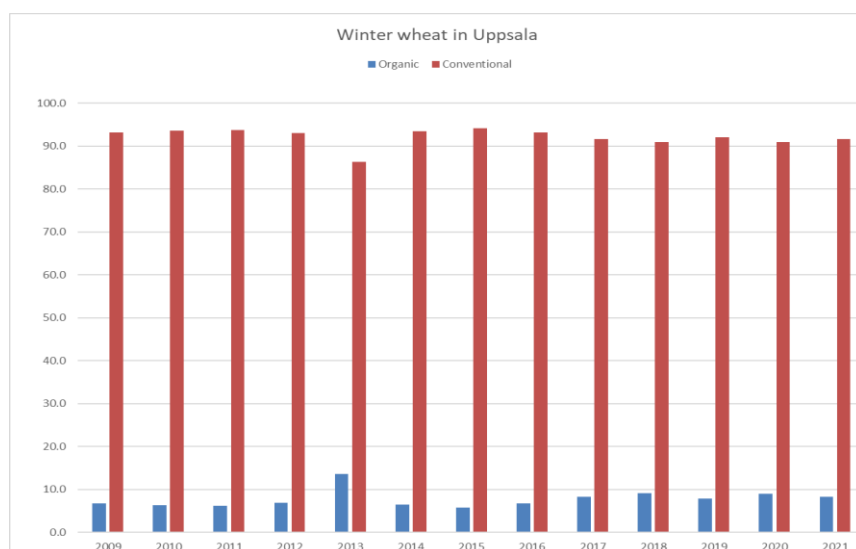
Share of total organically kept animals, comparison of number of animals in the year 2021 vs 2020, and number of organic companies

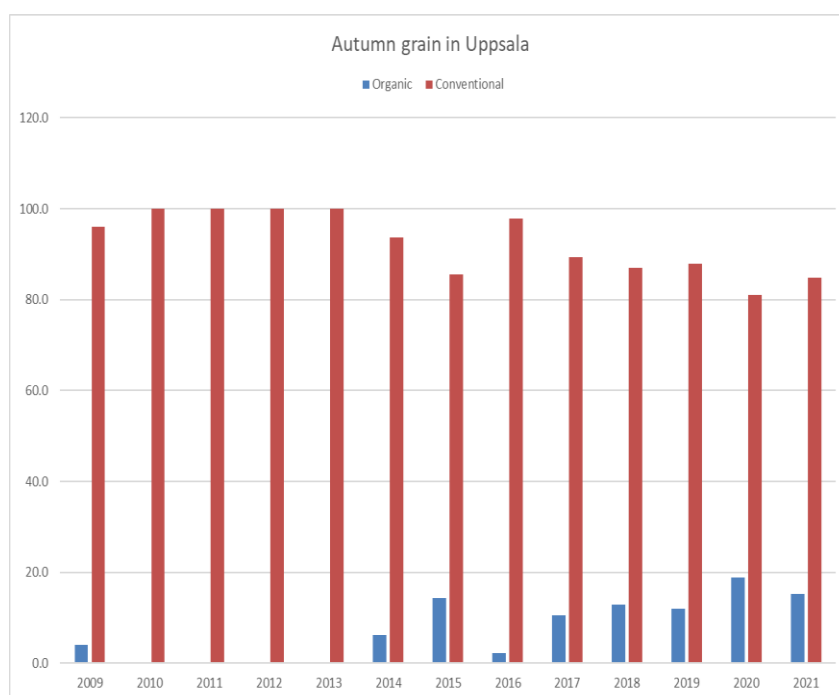
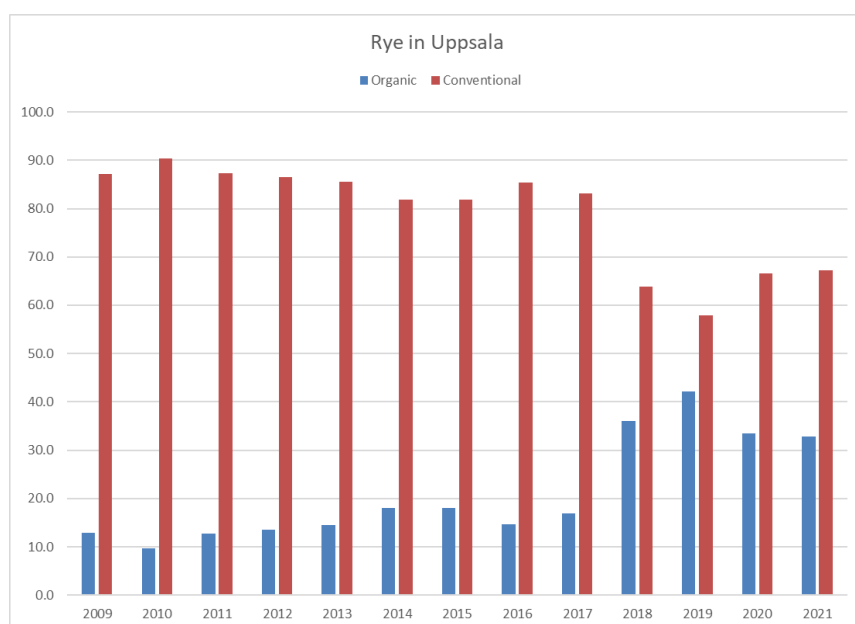
Animals	Share of total	Behaviour in number of animals in percentage vs 2020	Number of organic companies
Cows for milk production	36% (Increased 1% of the total share vs 2020)	Increase of 0.77%	33 (same as 2020)

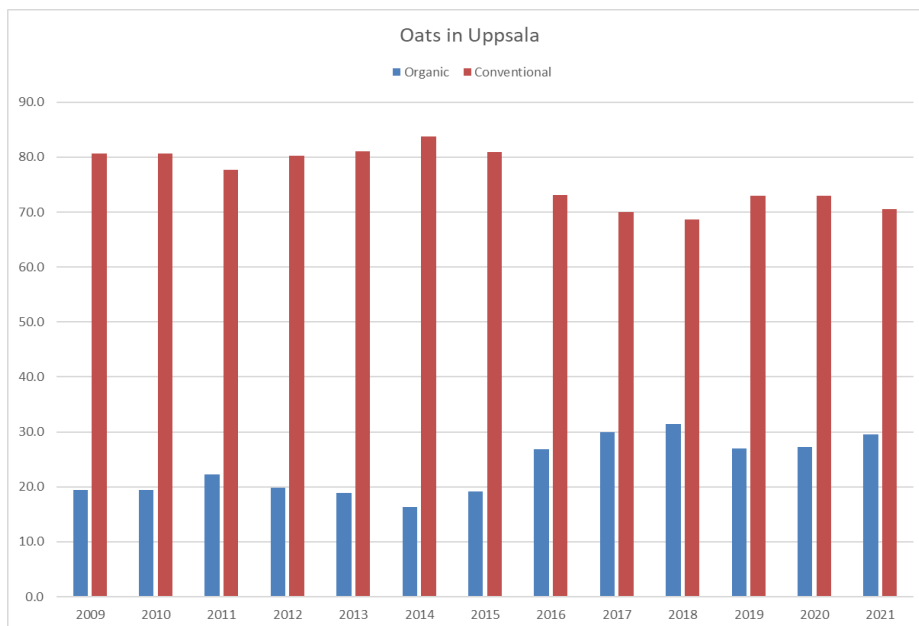
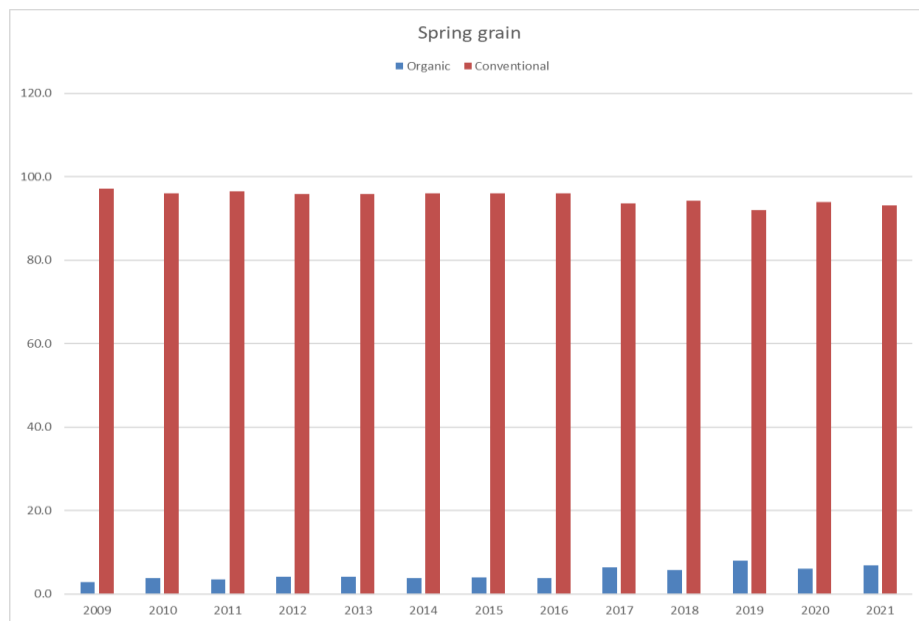
Cows for raising calves	36% (Decreased 3% of the total share vs 2020)	Decreased 2.5%	67 (decreased by 4 companies vs 2020)
Total cows	36% (Decreased 1% of the total share vs 2020)	Decreased 1.5%	100 (decreased by 4 companies vs 2020)
Heifers, bulls and steers	32% (same share as 2020)	Increased 2.7%	106 (decreased by 6 companies vs 2020)
Calves, under 1 year	28% (Decreased 1% of the total share vs 2020)	Decreased 2.4%	100 (decreased by 4 companies vs 2020)
Total cattle	32% (Decreased 1% of the total share vs 2020)	Decreased 0.4%	109 (decreased by 6 companies vs 2020)
Rams and ewes	15% (Increased 1% of the total share vs 2020)	Increased 2%	28 (decreased by 3 companies vs 2020)
Lamb	20% (Decreased 3% of the total share vs 2020)	Increased 12%	25 (decreased by 2 companies vs 2020)
Total sheep and lambs	18% (Decreased 3% of the total share vs 2020)	Increased 7.5%	28 (decreased by 3 companies vs 2020)
Boars for breeding	Only National data	Only National data	Only National data
Sows for breeding	None in 2021	111 animals in 2020	3 companies in 2020
Fatter pigs, 20 kg and above	8% (same as 2020)	No increment	5 (same as 2020)
Piglets, under 20 kg	None since 2018	None since 2018	None since 2018

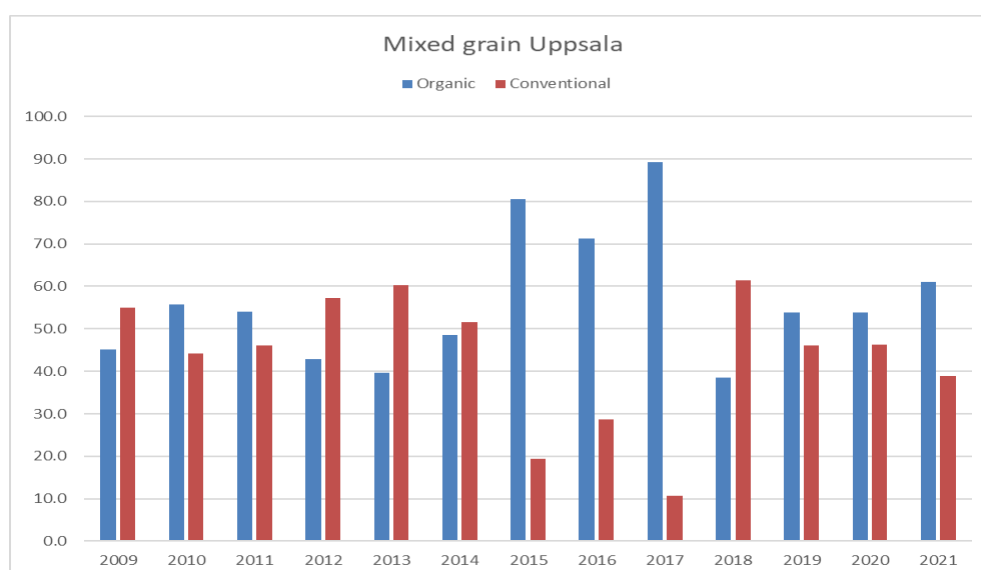
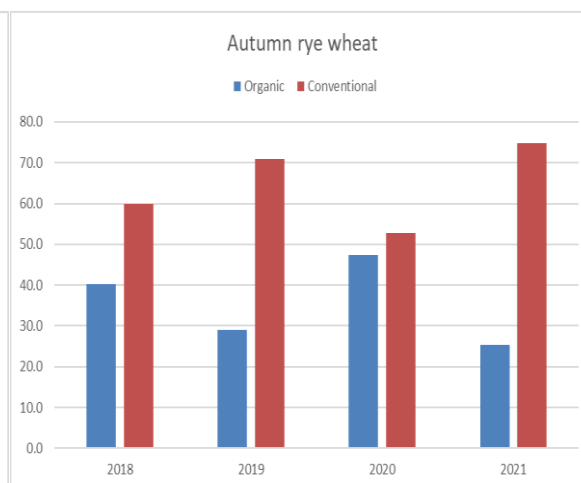
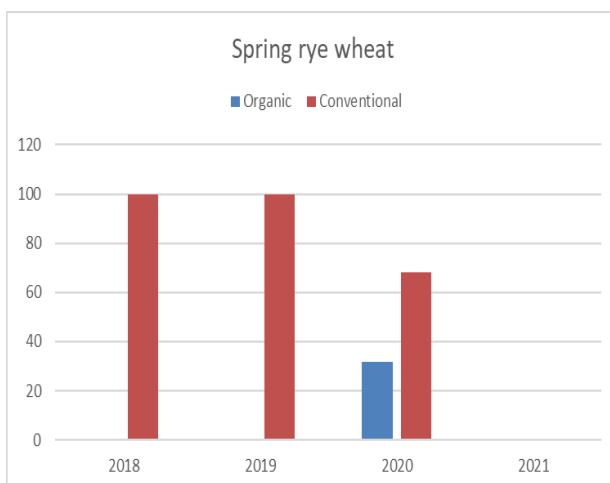
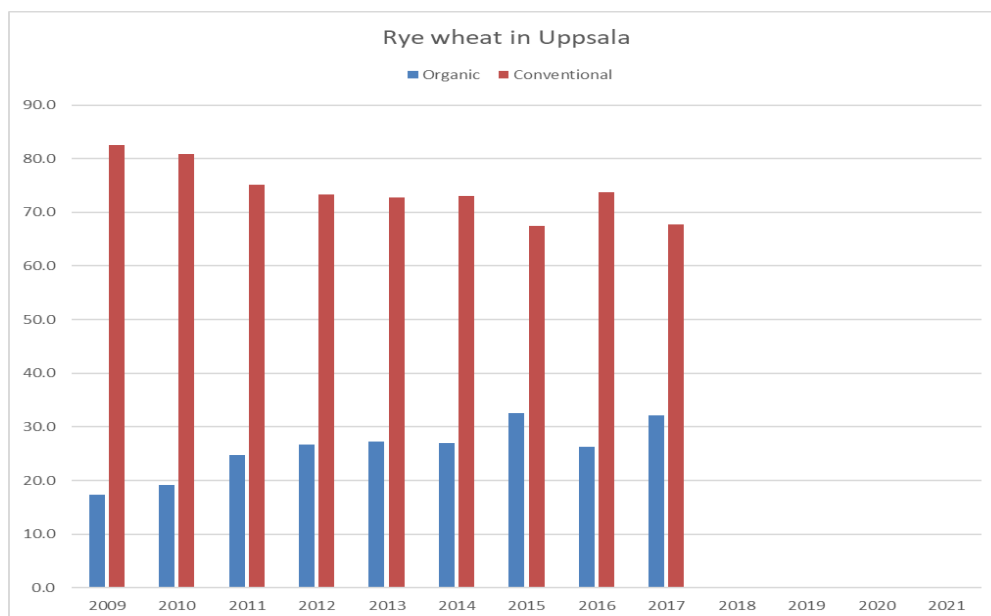
Total pigs	6% (Decreased 2% of the total share vs 2020)	Decreased 22%	
Laying hens	Only National data	Only National data	6 (same as 2020)
Broilers	Only National data	Only National data	Only National data

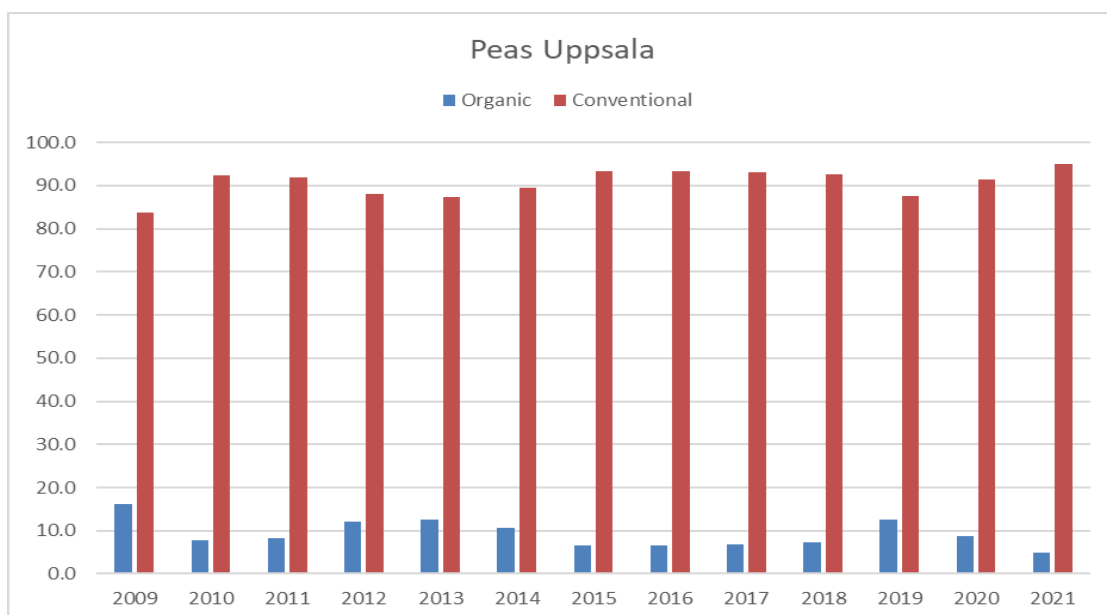
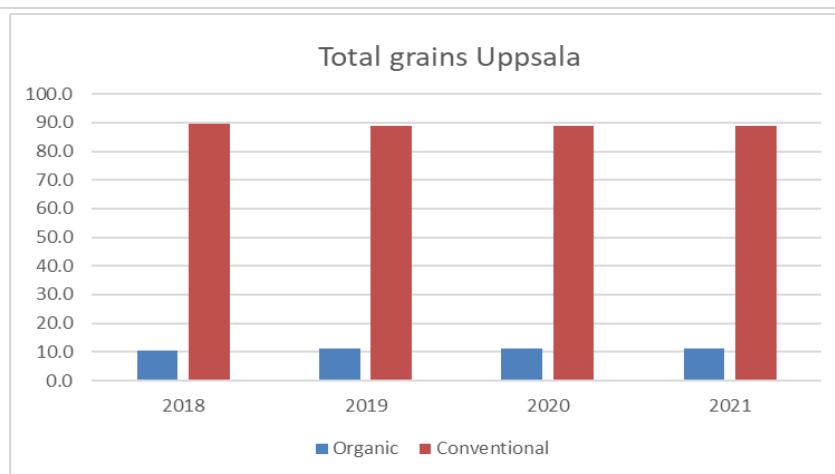
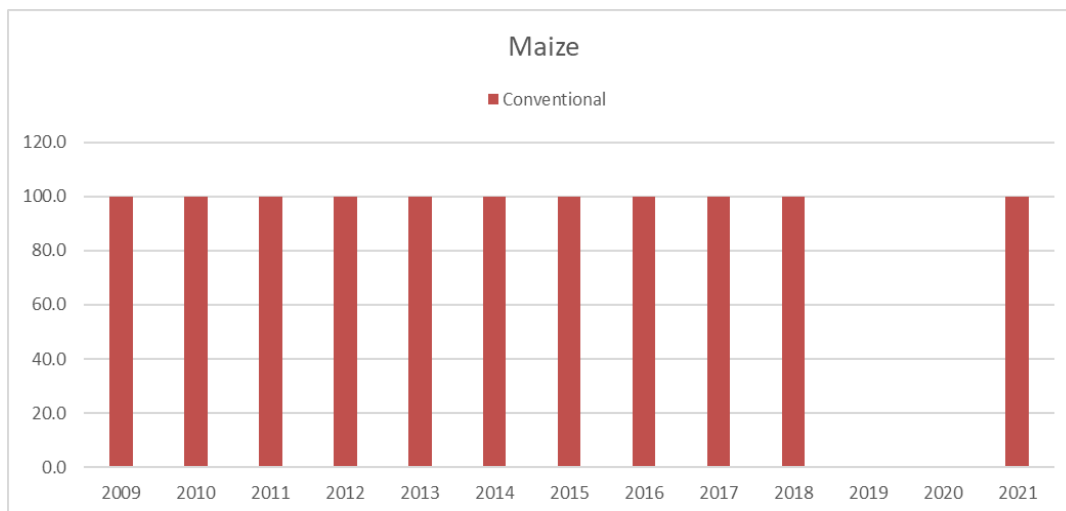
Appendix C: Comparison of crop area percent Organic vs Conventional in Uppsala County

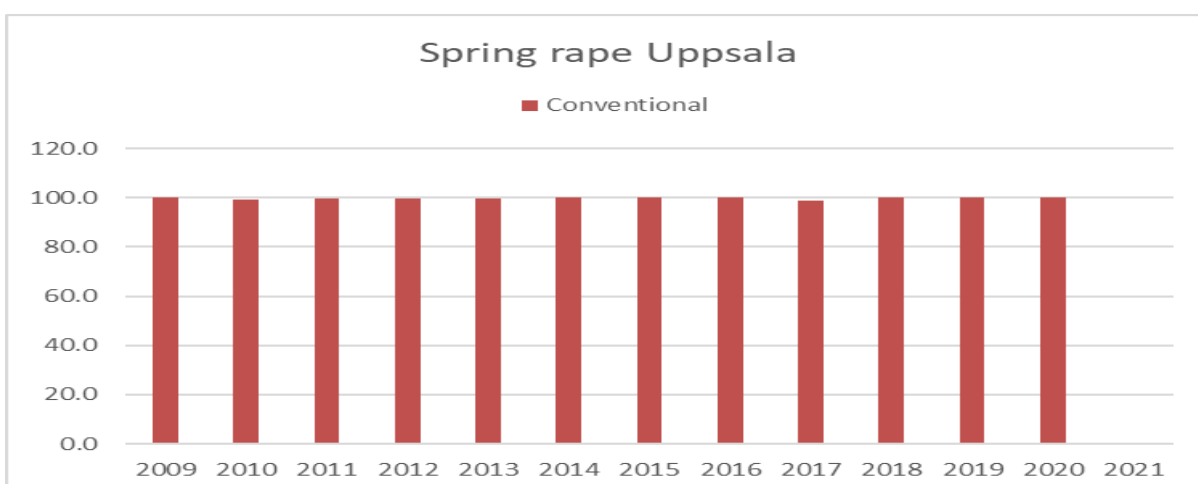
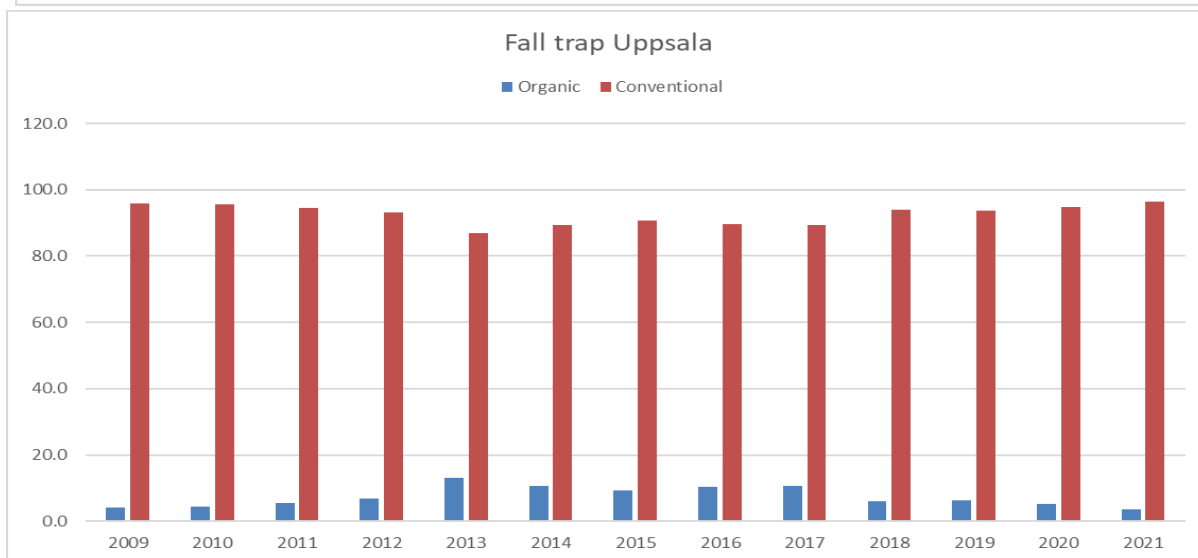
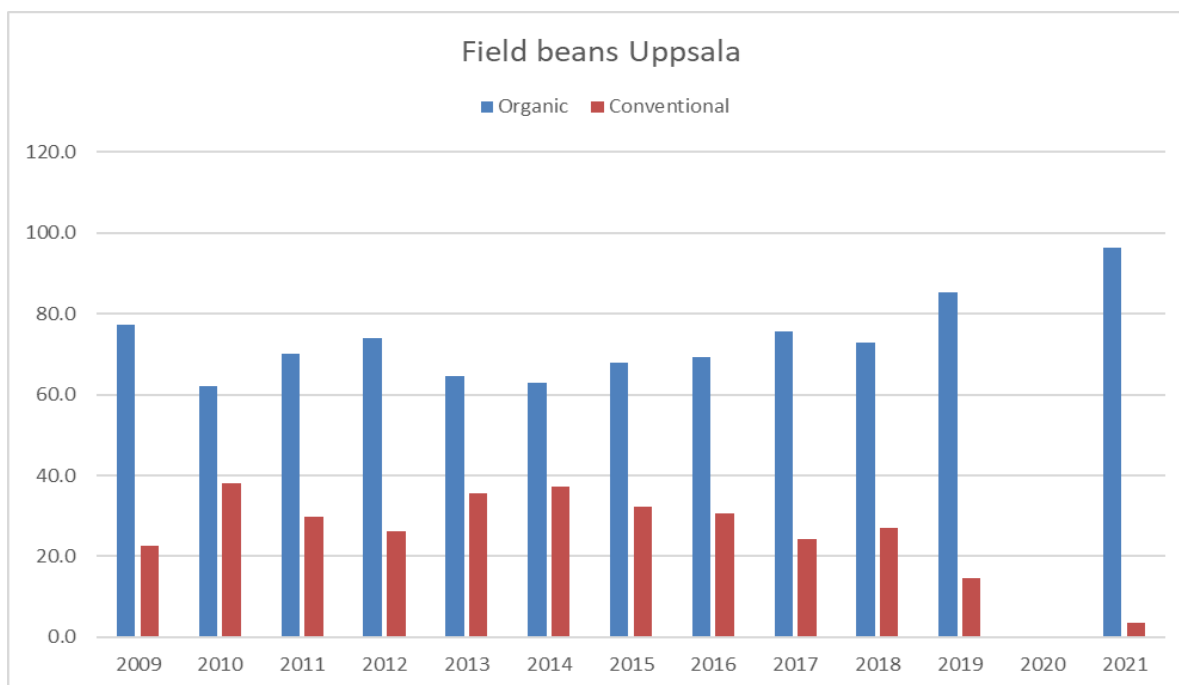


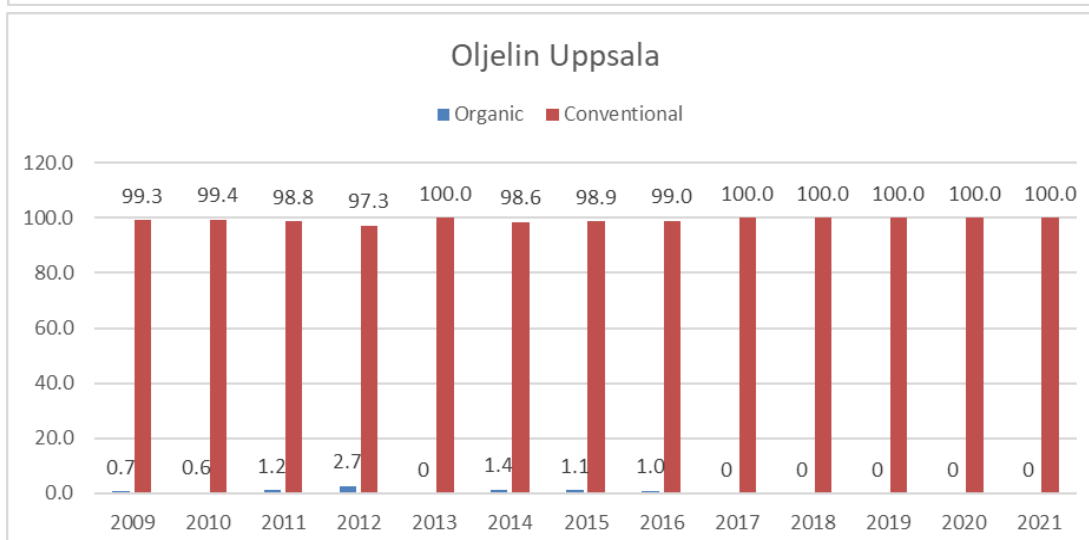
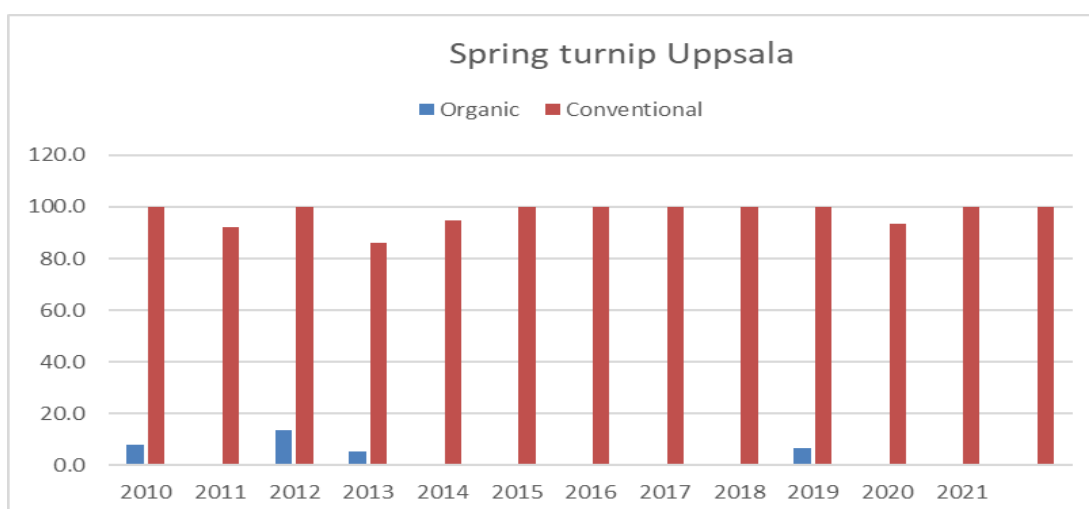
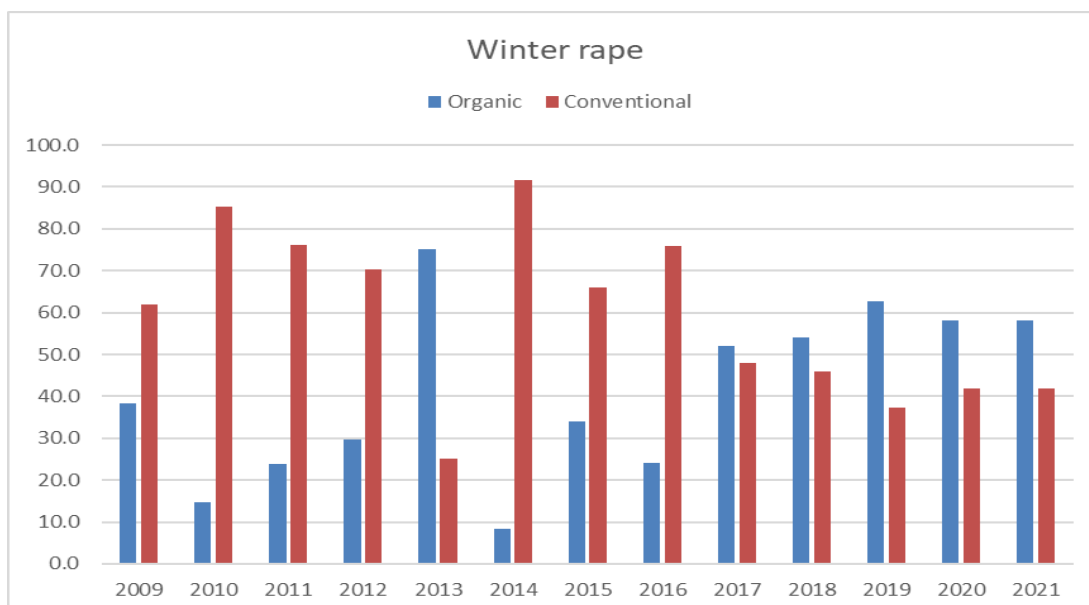






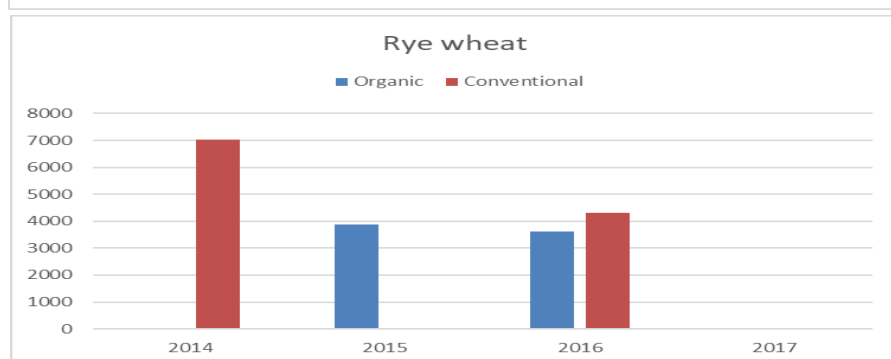
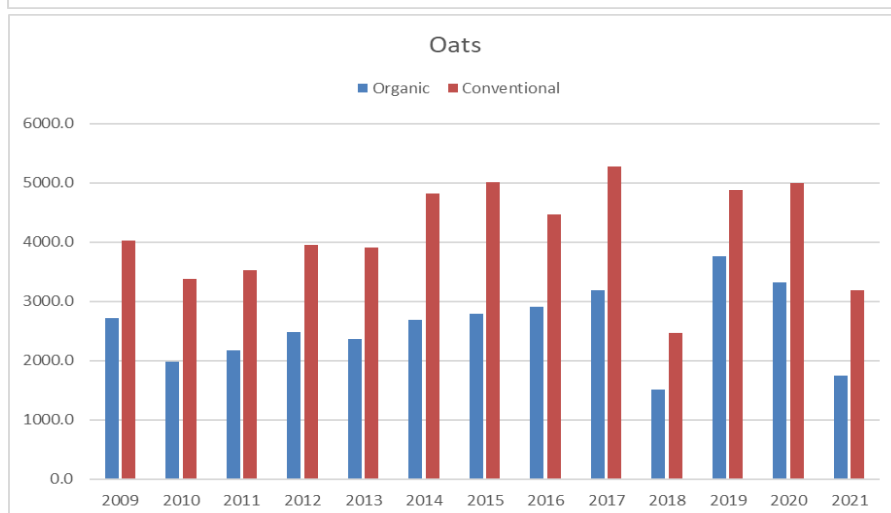
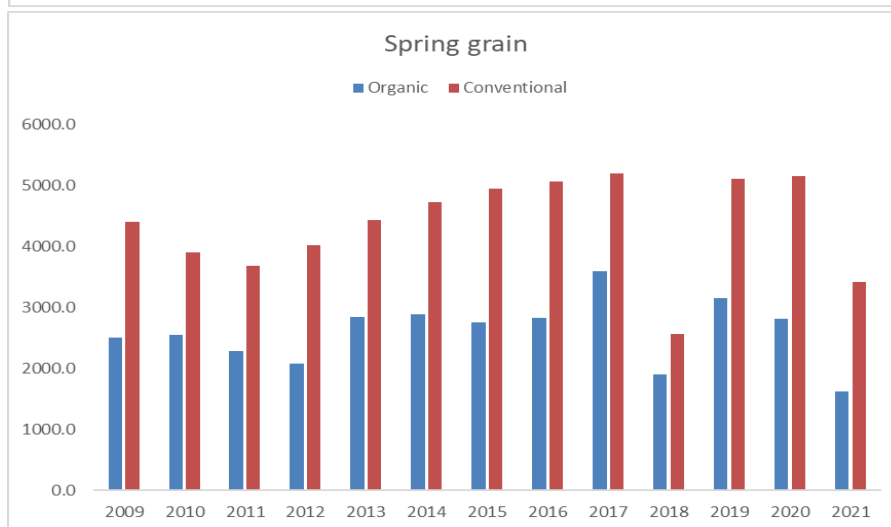
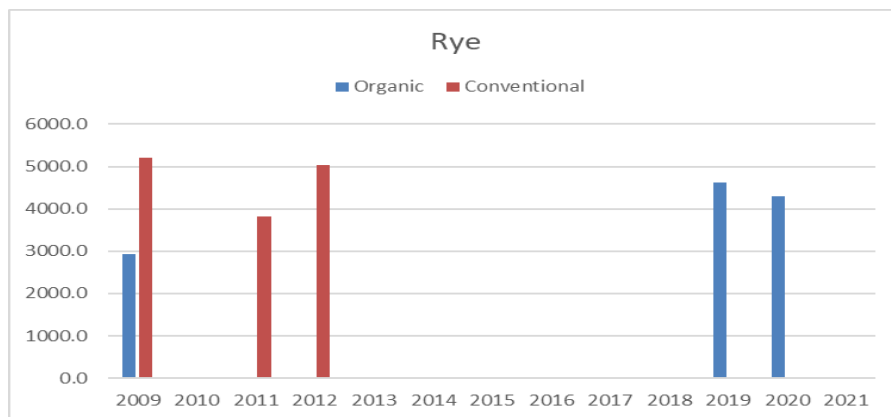


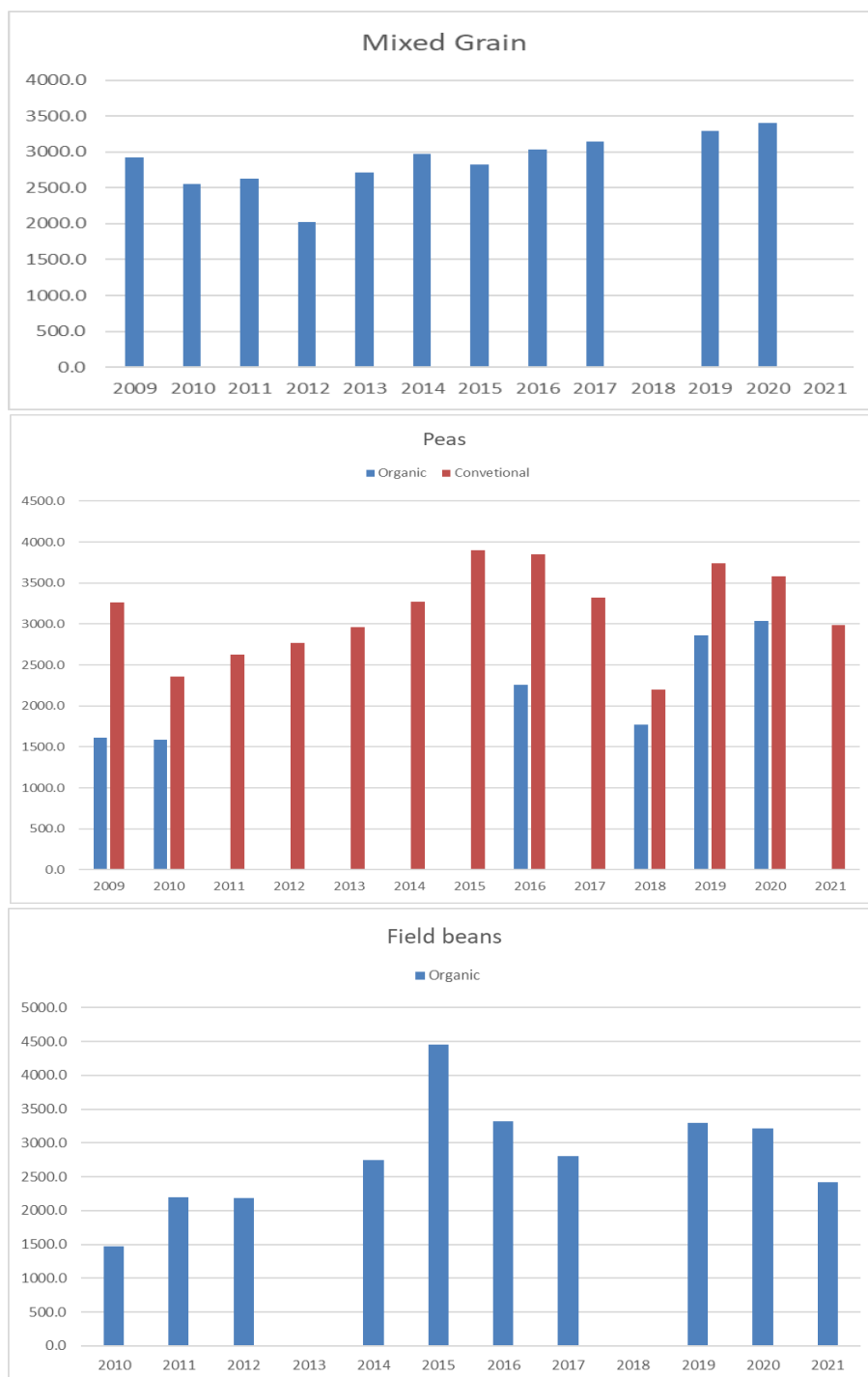


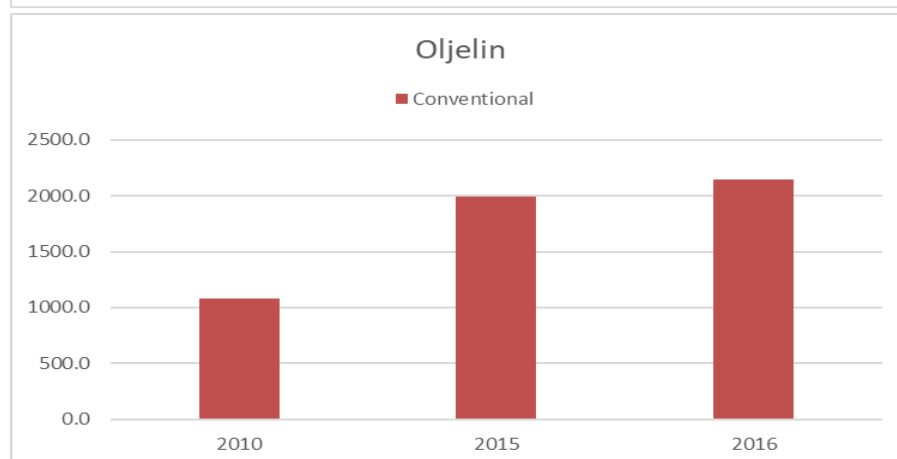
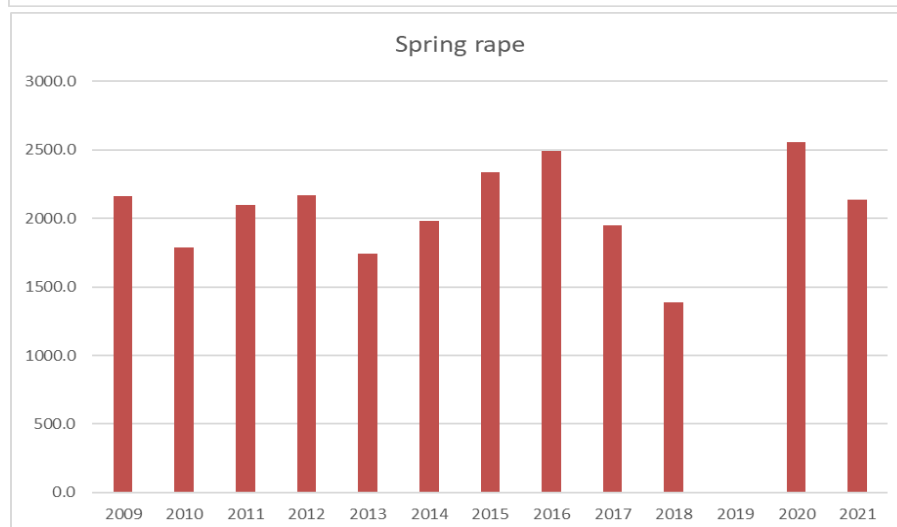
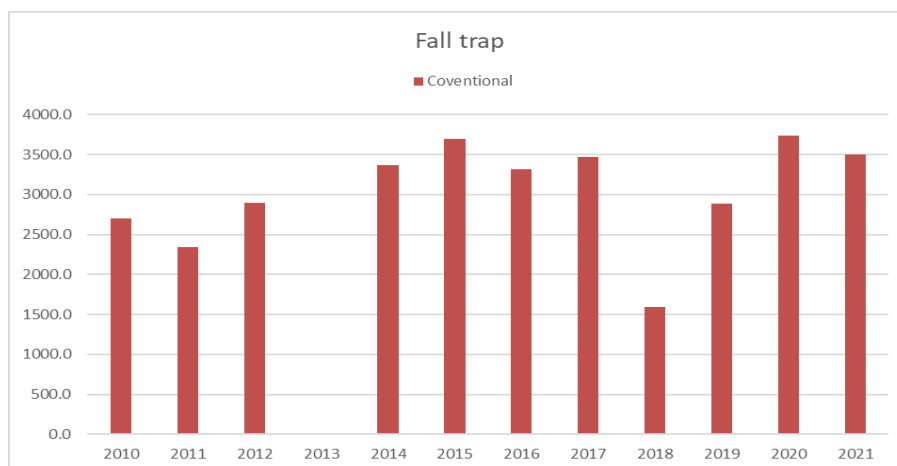












Appendix D: Production of fruits and vegetables in Uppsala County

	2017	2018	2019	2020	+/-
Cauliflower/ broccoli	49	41	55	25	Decreased
	390	310	390	230	Decreased
Onions	110	95	133	200	Increased
	2,330	1,770	2,440	7,790	Increased
Carrot	514	683	643	610	Decreased
	29,790	32,390	36,310	37,070	Increased
Lettuce	73	49	72	86	Increased
	420	240	320	1,040	Increased
Cabbage	153	119	136	108	Decreased
	2,420	3,930	4,750	4,620	Decreased
Other	415	348	482	481	Decreased
	7,050	5,870	9,096	9,420	Increased
Strawberries	79	239	136	96	Decreased
	210	600	380	340	Decreased
Blackcurrants	219	232	221	267	Increased
	300	280	260	150	Decreased
Apple	144	301	255	258	Increased
	1,720	2,900	1,660	2,540	Increased

